

Algebraic Curves and the Riemann-Roch Theorem

Yosime Flood

August 2024

Abstract

This thesis provides a detailed discussion of algebraic curves and the celebrated Riemann-Roch theorem (Theorem 8.0.1) using the classical tools of algebraic geometry. We begin by revisiting fundamental concepts, such as varieties, morphisms, and intersection numbers, before delving into the intricacies of Cremona transformations and blowups. After completing our discussion of blowups, we will prove the existence of nonsingular models, which allow for the rigorous treatment of divisors and $L(D)$ spaces. Through a detailed analysis of Riemann's inequality, canonical divisors, and ultimately, the Riemann-Roch theorem, we will establish the key results regarding the function spaces of algebraic curves. The ideas required in building the necessary machinery to prove the Riemann-Roch theorem are largely taken from "Algebraic Curves" [5], although we have extended the arguments found there to include a more detailed treatment of the case of fields with positive characteristic. In particular, we provide a detailed discussion of "terrible points" and a discussion of Kähler differentials without relying on the primitive element theorem. The thesis concludes with practical applications of the Riemann-Roch theorem. This includes a discussion towards the classification of elliptic curves, the derivation of the Hurwitz formula, and a comparison of the topological and algebraic genus, demonstrating its broad relevance and applicability.

Contents

1	Introduction	3
2	Algebraic Preliminaries	5
2.1	Varieties and Morphisms	5
2.2	Plane Curves	7
2.3	Rational Maps	14
3	Cremona Transformations	15
3.1	The Standard Cremona Transformation	15
3.2	Well-Placed Curves in Positive Characteristic	18
3.3	The Projective Space of Forms	21

4	The Blowup of a Curve	26
4.1	The Affine Blowup	27
4.2	The Projective Blowup	30
4.3	Nonsingular Models	32
5	Divisors and $L(D)$ Spaces	33
5.1	Divisors	33
5.2	$L(D)$ Spaces	36
6	Riemann's Inequality and the Genus of a Curve	38
6.1	Riemann's Inequality	38
6.2	The Genus Formula	42
7	Canonical Divisors	46
7.1	Kähler Differentials.	46
7.2	Canonical Divisors	50
8	Riemann-Roch Theorem	55
8.1	Proof of the Riemann-Roch Theorem	56
9	Applications of the Riemann-Roch Theorem	58
9.1	Linear Series	59
9.2	Hurwitz Formula	64
10	The Topological Genus	67
10.1	Analytic Structure	67
10.2	Euler Characteristic and Genus	69

1 Introduction

In the article “Theorie der Abel’schen Functionen” [1] Bernhard Riemann investigated the behavior of meromorphic functions defined on open subsets U of the Riemann sphere $\hat{\mathbb{C}}$. In particular interest to Riemann was the dimension of the space of meromorphic functions. In general, this space is infinite-dimensional; however, if we restrict the space to those meromorphic functions that do not have “too many” poles, this space becomes finite-dimensional. Riemann saw that the dimension was dependent on the topological structure of the open set U and that it grew roughly linearly in the number of poles that he allowed the functions to have. He could give a lower bound on this dimension, which may be viewed as a precursor to Riemann’s inequality 6.0.1 seen in Section 6. Riemann’s student, Gustav Roch, was later able to improve Riemann’s inequality by generalizing to Riemann surfaces embedded in $\mathbb{CP}^2$. He also found the exact dimension such function spaces should have, in which he improved Riemann’s Inequality to the Riemann-Roch theorem [2]. Although Riemann and Roch discussed the context of this theorem in a purely analytic context, many of the ideas can be extended analogously to algebraic curves over other fields. The first to do this were Dedekind and Weber [3], who not only generalized the results of Riemann and Roch but also corrected the proof.

In its most modern form, the Riemann-Roch theorem has been generalized to the Grothendieck–Riemann–Roch theorem, which is stated in the modern language of schemes and cohomology. However, in this thesis, we refrain from using the most modern tools and follow an elementary approach.

Following the classical approach, we will also not discuss the generic points that occur when extending the classic theory to the more general language of spectra of rings.

The ideas we follow can be attributed to the influence of Max Noether and Alexander von Brill [4]; although for the execution, we have almost exclusively followed Fulton’s “Algebraic Curves” [5]. However, we have refined some of the arguments for the case where the base field has nonzero characteristic. In particular, our discussion in Section 3 on quadratic transformations has extended the arguments seen in Fulton to include a detailed discussion about terrible points. In our discussion on differential forms, we have also discussed the case $\text{char}(k) = p$ in more detail, such that we must not rely on the theorem of the primitive element to prove theorems such as 7.1.5.

The primary exception to following Fulton’s book occurs in Section 10, where we follow Kendig [6] and Lang [7]. Another exception occurs in Section 9, where we show a few applications of the Riemann-Roch theorem.

The key objects that we study are algebraic curves over some algebraically closed field k (which will be defined properly in Section 2). Taking inspiration from differential geometry, we seek to study these objects by the functions defined on them; for example, when $k = \mathbb{C}$, we may be interested in the space of holomorphic functions on the complex algebraic curve $\hat{\mathbb{C}} = \mathbb{CP}^1$. Considering Liouville’s theorem, we know these are only the constant functions. In fact, an analog of Liouville’s theorem will hold for any projective algebraic curve,

as we will see in Section 2. This consideration shows that if we hope to investigate any interesting properties of our algebraic curve, we need to allow the considered functions to have at least some poles, which in an analytic context corresponds to considering meromorphic functions rather than entire functions. To do this, we will introduce the notion of a divisor D on a curve, which will keep track of where and of what nature our function's singularities may be. We will associate a function space called $L(D)$ to each divisor. This space consists of (meromorphic-like) functions on our algebraic curve, which only have poles at those points specified by D . The Riemann-Roch theorem will then give the dimension of $L(D)$ as a function of the "size" of D , the topological structure of the curve in consideration, and the space of differentials on the curve. The theorem states:

$$\dim_k(L(D)) = \deg(D) + 1 - g + \dim_k(L(W - D)). \quad (1)$$

We will explain what each of the terms in the above equation precisely means in Sections 5 and 7. The theorem effectively allows us to understand how meromorphic functions (in a generalized sense of meromorphic) behave on algebraic curves.

We will now provide a brief overview of the sections of this thesis. In Section 2, we will recall facts about algebraic curves and varieties in general, which are foundational to this thesis. We pay particular attention to some subtleties which can occur when the base field over which we work has nonzero characteristic. The main goal of Sections 3 and 4 is to show that an arbitrary algebraic curve may be transformed birationally to a unique smooth curve. This is done in two steps: In Section 3, we show that an arbitrary algebraic curve may be transformed to a plane curve which has more "manageable" singularities called ordinary singularities; in Section 4, we then show that planar curves with only ordinary singularities may be birationally transformed to smooth curves using the blowup. This provides the necessary tools for proving the existence of non-singular models.

In Section 5, divisors and $L(D)$ spaces are introduced. These objects allow for formally quantifying the meaning of a function having specified poles at specified places on the algebraic curve.

In Section 6, we proceed to prove the first results on $L(D)$ spaces, such as Riemann's inequality. These results provide a first view of how large the space of functions on an algebraic curve is when only a select number of poles at a select number of places are allowed. Riemann's inequality will also allow the formulation of the genus in a purely algebraic context; in particular, it allows for a justified generalization of the genus in topology to curves over more general fields.

In Section 7, we discuss canonical divisors and introduce algebraic differentials. The use of these objects is the crucial contribution of Roch, which allowed Riemann's inequality to be refined to the Riemann-Roch theorem. In Section 8, we can finally prove the Riemann-Roch theorem.

Throughout this thesis, many terms, such as the genus of a curve or the dimension of a variety, are used but phrased in a purely algebraic context. It is natural to ask whether these notions correspond to their counterparts in topology and geometry when the base field is $\mathbb{C}$. Indeed, this is the case, and we will prove this in Section 10. In Section 9, we will see the general applicability of the Riemann-Roch theorem. We will use the theorem to make the first step towards classifying elliptic curves, to prove the Weierstrass formula, and to prove the Hurwitz formula.

2 Algebraic Preliminaries

This thesis will rely on many basic facts from commutative algebra and algebraic geometry. In this section, we set the stage for the rest of the thesis by recalling a few definitions and results. In this section exclusively, we will only prove those results that may not occur in an introductory course in commutative algebra or algebraic curves.

The main object of interest in this thesis is the algebraic curve and the concepts that come along with it. Historically, the concept of an algebraic curve arose by studying those one-dimensional manifolds (over $\mathbb{R}$ or $\mathbb{C}$), which may be described as the solutions to polynomial equations. The powerful theory developed in geometry to study these solution spaces was then generalized to study the solution spaces to systems of polynomial equations over other fields and rings. Perhaps one of the most famous examples is the study of the Fermat curve, which asks for the rational solutions to the equation $X^n + Y^n = Z^n$. This process of generalizing the notions from geometry to purely algebraic notions led to the definition of a variety and, later, the definition of a scheme.

2.1 Varieties and Morphisms

In this thesis, an algebraic curve will be a one-dimensional variety over an algebraically closed field k . Let us fix k as such a field for the rest of this thesis. For the sake of simplicity, we will not take the most general definition of variety. Namely, we will consider only quasi-multiprojective varieties, as seen below.

Definition 2.1.1. Let $n_1, \dots, n_k, m$ be positive integers, and let $\mathbb{P}^{n_i}$ be projective n_i -spaces over k . Furthermore denote by $\mathbb{A}^m$ the space k^m and by X the space $\mathbb{P}^{n_1} \times \dots \times \mathbb{P}^{n_k} \times \mathbb{A}^m$. A closed set of X is defined to be a set of the form $V(I)$ where I is a semi-multi-homogeneous ideal in the ring

$$\Gamma := k[X_{1,0}, \dots, X_{1,n_1}, X_{2,0}, \dots, X_{k,n_k}, Y_1, \dots, Y_m],$$

i.e., an ideal that has a set of generators that are homogeneous in each of the variables in their respective projective spaces (possibly of different degrees).

Remark. The prefix “semi” in “semi-multi-homogeneous ideal” alludes to the fact that the restrictions we placed on the ideal I were only on the variables

corresponding to the projective spaces $\mathbb{P}^{n_i}$. The variables corresponding to $\mathbb{A}^m$ were not restricted in any way.

Remark. Strictly speaking, considering X of the form $\mathbb{P}^N \times \mathbb{A}^m$ is sufficient since the spaces above admit closed embeddings into such spaces via the Segre map.

The sets $V(I)$ form the closed sets of a topology on X called the Zariski topology on X . If X is of the simple form $\mathbb{A}^m$, we call these sets affine; and if X has the form $\mathbb{P}^n$, we call them projective. More generally, a variety that is isomorphic (in the sense defined below) to an affine (or projective) variety is called an affine (or projective) variety. Assuming the notation from the above definition, we can define the notion of variety used in this thesis.

Definition 2.1.2. A closed variety is an irreducible closed subset of some space X of the above form. A variety is an open subset of a closed variety.

Given a variety V , we would like to assign its function field $K(V)$ to it. To any open subset of the variety, we are also interested in those functions in $K(V)$ which do not have poles on this set. More precisely, we would like to define its sheaf of regular functions; however, we will not appeal to sheaf theory here. For a closed variety V in X , the set of polynomials in Γ (as above) which vanish on V form a semi-multi-homogeneous ideal I in Γ . We can then define the multi-homogeneous coordinate ring R_V of V as Γ/I . The elements in R_V do not yet define well-defined functions on V since their values are not invariant under scalar multiplication in each projective space. However, they allow us to define the function field $K(V)$ of V as those elements in R_V 's field of fractions, which may be represented as a quotient of multi-homogeneous polynomials of equal multi-degree. For an open variety V , we consider its (Zariski) closure $\bar{V}$. Since the closure of an irreducible set is irreducible, $\bar{V}$ is a closed variety. This allows us to define $K(V) := K(\bar{V})$.

If we are given a point $P \in V$, we also define the ring $\mathcal{O}_P(V)$ as

$$\mathcal{O}_P(V) := \{x \in K(V) \mid \text{where } x \text{ has representation } f/g \in K(V), \text{ with } g(P) \neq 0\}.$$

With this definition, every element in $\mathcal{O}_P(V)$ takes on a unique, well-defined value in k at P , and $\mathcal{O}_P(V)$ is a local ring with maximal ideal consisting of those functions which vanish at P . Indeed, the set of elements in $\mathcal{O}_P(V)$, which vanish at P , form an ideal m . Furthermore, if an element $x \in \mathcal{O}_P(V)$ does not lie in m , then x admits a representation $x = f/g$ where $g(P) \neq 0 \neq f(P)$, thus $x^{-1} = g/f$ also lies in $\mathcal{O}_P(V)$. From this, it follows that every element not in m is a unit, so $\mathcal{O}_P(V)$ is a local ring. We will call this ring the local ring of V at P .

For any open set $U \subseteq V$ we can define the global sections of U as $\Gamma_V(U) := \bigcap_{P \in U} \mathcal{O}_P(V)$. When V is affine, up to isomorphism of varieties (defined below), V is given by a set $V(I) \subseteq \mathbb{A}^m$ where I is an ideal in $k[X_1, \dots, X_n]$. Since V is irreducible, one may show that I is primary. One may then show that $\Gamma_V(V)$, as we have defined it here, is isomorphic to $k[X_1, \dots, X_n]/\sqrt{I}$. On the contrary, if V is projective, it is known that $\Gamma_V(V) = k$. Equivalently, one may state that

no nonconstant global sections on projective varieties exist. A proof of these facts can be found in [5].

Definition 2.1.3. Let V and W be two varieties. A morphism of varieties is defined as a continuous map of topological spaces $f: V \rightarrow W$ such that if U is an open of W and $z \in \Gamma_W(U)$ then $f^b(z) := z \circ f$ is an element of $\Gamma_V(f^{-1}(U))$.

The dimension of a variety is defined as the transcendence degree of its function field over k . In this thesis, we are only interested in one-dimensional varieties.

Definition 2.1.4. An algebraic curve is a variety of dimension one. An affine (respectively projective) curve is an affine (respectively projective) variety of dimension one. An affine (respectively projective) plane curve is an algebraic curve that is a closed subset of $\mathbb{A}^2$ (respectively of $\mathbb{P}^2$).

Affine plane curves are given by sets of the form $V(F)$ where $F \in k[X, Y]$ is a primary polynomial. Since $V(F) = V(\sqrt{(F)})$, when we refer to an algebraic curve, we assume F to be square-free and thus, in this case, irreducible. Similarly, projective plane curves will be represented by sets of the form $V(F)$ where $F \in k[X, Y, Z]$ is an irreducible form.

2.2 Plane Curves

Many useful insights about plane curves can be used to gain insight into general algebraic curves. Take, for example, an affine plane curve; its coordinate ring is given by a quotient ring $K[X, Y]/I$, and thus (relabeling variables if needed), its function field may be written as $k(x, y)$ where x is transcendental over k and y is algebraic over $k(x)$. As it turns out, the function field of any algebraic curve will have this simple form.

Lemma 2.2.1. *Let V be an algebraic curve and $K(V)$ its function field. Then $K(V)$ may be written in the simple form $k(x, y)$, where x is transcendental over k and y is algebraic and separable over $k(x)$.*

Proof. By definition $k(V) = k(x, y_1, \dots, y_n)$ where x is transcendental over k and the y_i are algebraic over $k(x)$. If $\text{char}(k) = 0$, the theorem of the primitive element ensures that $k(x)(y_1, \dots, y_n) = k(x, y)$ for some primitive element y of the given field extension. It remains to show the case $\text{char}(k) = p > 0$. We use the fact that the field k is algebraically closed. It suffices to show that there exists an element $x \in k(V)$, which is transcendental over k , such that $K(V)/k(x)$ is a separable extension. Indeed, this suffices since separable extensions admit (separable) primitive elements.

We induct over n . If $n = 0$, then $k(V) = k(x)$ and there is nothing to prove, so assume $n > 0$. Since $k(V)$ has transcendence degree one over k , we have a polynomial $f \in k[X, Y_1, \dots, Y_n]$ such that $f(x, y_1, \dots, y_n) = 0$. Without loss of generality, let it be irreducible, and let all the y_i also be transcendental over k (or else they lie k). Then not all variables $X, Y_1, \dots, Y_n$, occur only in p 'th

powers in f . Indeed, if this were not the case, f would be contained in the ring $k[X^p, Y_1^p, \dots, Y_n^p]$, which, using the Frobenius morphism, would yield a factorization of f as

$$f(X, Y_1, \dots, Y_n) = \tilde{f}(X^p, Y_1^p, \dots, Y_n^p) = g(X, Y_1, \dots, Y_n)^p,$$

where g is the polynomial whose coefficients correspond to p^{th} roots of those of $\tilde{f}$. Without loss of generality, suppose Y_n does not only occur in powers of p . Then the minimal polynomial of y_n over $k(x, y_1, \dots, y_{n-1})$ is $f(x, y_1, \dots, y_{n-1}, Y_n)$ and y_n is separable over $k(x, y_1, \dots, y_{n-1})$, since the derivative of f in Y_n is nonvanishing. We may apply the induction hypothesis to $k(x, y_1, \dots, y_{n-1})$ to ensure we have an element $\tilde{x}$ which is transcendental over k and such that $k(x, y_1, \dots, y_{n-1})$ is a separable extension over $k(\tilde{x})$. By the transitive property of separable extensions, it follows that $k(x, y_1, \dots, y_n)$ must be separable over $k(\tilde{x})$. It thus also admits a (separable) primitive element $\tilde{y}$, which yields $k(V) = k(\tilde{x}, \tilde{y})$. $\square$

Now that we have seen the general structure in which the function fields of algebraic curves can occur, we collect a few famous facts about plane curves, which we will use throughout this thesis. First, we recall the definition of a smooth point on a curve.

Definition 2.2.2. Let $C = V(F)$ be an affine plane curve defined by an irreducible polynomial F . denote by F_X and F_Y the partial derivatives of F . A point $P \in C$ is called singular if $F_X(P) = F_Y(P) = 0$; otherwise, P is called a smooth point of C .

If $C = V(F) \subseteq \mathbb{P}^2$ is a projective plane curve, a point $P \in C$ is called singular if $F_X(P) = F_Y(P) = F_Z(P) = 0$ (recall our convention that F is irreducible). Equivalently, we may say that a point is singular on a projective curve if it is singular on some affine representative. One may show that, for plane curves, a point P is smooth on C if and only if its local ring $\mathcal{O}_P(C)$ is a discrete valuation ring of $K(C)$ containing k (see for example [5]). For a general algebraic curve, we define a point as smooth if its local ring is a discrete valuation ring of its function field containing k . Recall from commutative algebra that an order one element of a discrete valuation ring is called a uniformizing parameter and that if $k \subseteq R$ and R is a discrete valuation ring with field of fractions K with uniformizing parameter t , then every element in K may be identified with a unique power series in t .

As we will see, various types of singularities can appear on an algebraic curve. We begin our study of singularities by introducing the concept of multiplicity. Intuitively, the multiplicity of a point on a curve describes how often the curve goes through this point.

Definition 2.2.3. Let $F \in k[X, Y]$ be a nonzero polynomial and $P = (0, 0)$. Suppose F is given as $F = F_r(X, Y) + \dots + F_n(X, Y)$ where F_i is a form of degree i $F_r \neq 0$. The multiplicity of F at P , denoted by $m_P(F)$, is defined as $m_P(F) := r$. If C is an affine plane curve given by an irreducible form F , we define the multiplicity $m_P(C)$ as $m_P(C) := m_P(F)$. For a general point $Q \in \mathbb{A}^2$ choose the

translation $T = (T_1, T_2): \mathbb{A}^2 \rightarrow \mathbb{A}^2$ which maps P to Q . We define the multiplicity $m_Q(F)$ as the multiplicity $m_P(F^T)$ where $F^T = F \circ T = F(T_1(X, Y), T_2(X, Y))$.

For a projective plane curve, we may simply define the multiplicity of a point as the multiplicity of that point on any affine representative of the curve. Equivalently for $P = [0 : 0 : 1]$ we may write the defining form of C as

$$F = F_r(X, Y)Z^{n-1} + \cdots + F_n(X, Y)Z^0, F_r \neq 0,$$

and then obtain the multiplicity as the unique r obtained from the above representation.

The first question one may ask is how many singular points an algebraic curve may have. Taking inspiration from the classical theory in $\text{char}(k) = 0$, it seems there should be only finitely many such points. We will show that this result holds for plane curves. Our brief review of birational geometry later in this section will allow us to extend this to any algebraic curve. Furthermore, since $\mathbb{P}^2$ is covered by affines, it suffices to prove the result for affine curves since we may simply dehomogenize the defining form in the different canonical affines, count the singularities, and add them up.

Lemma 2.2.4. *Let $C = V(F)$ be an affine plane curve defined by an irreducible polynomial F . Then C has only finitely many singularities.*

Proof. The singular points of C are given by the algebraic set $S := V(F, F_X, F_Y)$. Since F is irreducible and F_X and F_Y each have lesser degree than F , they are either zero or coprime to F . Suppose at least one of the partial derivatives is nonzero. In that case, S becomes a proper algebraic subset of a curve; hence, a finite collection of points (by considering the Krull dimension of the coordinate ring). Thus, it suffices to show that F_X and F_Y cannot both be zero. In case $\text{char}(k) = 0$, having $F_X = 0 = F_Y$ implies that neither X nor Y occurs in F , which means F is a constant, contradicting the fact that C is an algebraic curve. In the case of positive characteristic, having vanishing derivatives implies that the variables X and Y occur only in powers of p , hence $F \in k[X^p, Y^p]$. This implies that F can be written as $F^*(X^p, Y^p)$, which yields a factorization $F = G(X, Y)^p$, where G is the polynomial whose coefficients are p 'th roots of those coefficients that F^* has. We conclude that F is reducible, which is a contradiction. $\square$

Remark. Because factoring forms in $k[X, Y, Z]$ is equivalent to factoring arbitrary polynomials in $k[X, Y]$, the above proof can also be used to show the statement for projective plane curves. For a general algebraic curve, use Lemma 2.2.1 to obtain a birational morphism (defined later) to a plane curve and notice that open subsets on curves simply look like the original curve with finitely many points removed.

The first part of this thesis will deal with removing singularities from a given curve. This process can differ greatly depending on what type of singularity we are dealing with. Generally, these singularities will fall into two categories: ordinary and nonordinary. Heuristically, the difference between these two types

of singularities lies in the tangents that we may place at our curve; at an ordinary singularity, our curve will have pairwise distinct tangents, and at a nonordinary singularity, we may have tangents occurring with multiplicity.

Definition 2.2.5. Let $C = V(F)$ be an affine curve and let $P = (0, 0) \in C$. Suppose we may write F as $F_r(X, Y) + \cdots + F_n(X, Y)$ where F_i is a form of degree i and $F_r \neq 0$. Since F_r is a form in two variables over an algebraically closed field, it factors as

$$F_r = \prod_{i=1}^r (\alpha_i X + \beta_i Y).$$

Suppose $r \geq 2$. We call P an ordinary singularity of C if, in the above factorization, F_r is square free. Otherwise, we call P a nonordinary singularity. If Q is any point in $\mathbb{A}^2$, let T be the affine translation, which maps P to Q . We call Q an ordinary singularity (a nonordinary singularity, respectively) if P is an ordinary singularity (a nonordinary singularity, respectively) on F^T .

The factors that occur in F_r are called the tangents to F at P , and the exponents in which they occur are called their multiplicities. For a general point Q , the tangents of F at Q are given by shifting the tangents that occur for F^T at P , by T .

Once again, we have two equivalent ways of generalizing these notions to projective curves. The first way is to take some affine representative, which contains the point in question, and then consider the tangents of the affine curve. By homogenizing these tangents, we obtain tangents of the original curve. The second way is to choose coordinates such that $Q = [0 : 0 : 1]$ and then write F as

$$F = F_r(X, Y)Z^{n-r} + \cdots + F_n(X, Y)Z^0, F_r \neq 0,$$

whereafter, we could take the factors that occur in F_r .

Let us consider an example. Consider the curve given by the set $V(Y^2Z - X^3) \subseteq \mathbb{P}^2$, which is illustrated below.

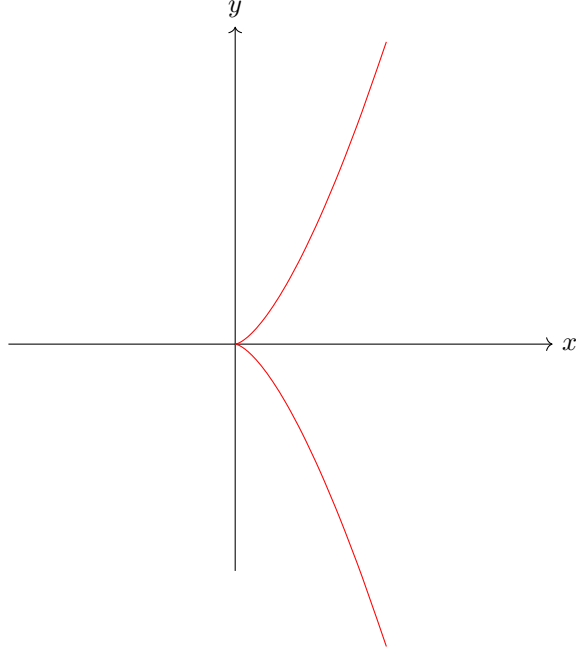


Figure 1: A curve with a cusp.

This curve has multiplicity two at $P = [0:0:1]$ since dehomogenizing $Y^2Z - X^3$ at Z yields the polynomial $Y^2 - X^3$, which has lowest term Y^2 of degree two. Furthermore, the lowest term is Y^2 , which factors as $Y \cdot Y$. Thus $P = [0:0:1]$ is nonordinary singularity of F and Y is a double tangent of F .

In a sense, the tangents of a curve at a point are those lines that intersect our curve in some higher order. To make this precise, we need to use the concept of an intersection number, which we will now introduce. For proof of the below statements, see [5].

Definition 2.2.6. Let F and G be two nonconstant polynomials in $k[X, Y]$ and let $P \in \mathbb{A}^2$ be an affine point. The intersection number of F and G is defined as

$$I(P; F, G) := \dim_k(\mathcal{O}_P(\mathbb{A}^2)/(F, G)).$$

It is the unique number which satisfies the following list of properties:

- 1) $I(P; F, G)$ is a nonnegative integer whenever F and G do not share a common factor which vanishes at P . Otherwise, it is ∞ .
- 2) $I(P; F, G)$ is zero if and only if P does not lie in $V(F) \cap V(G)$.
- 3) $I(P; F, G)$ is invariant under affine changes of coordinates.
- 4) $I(P; F, G)$ is symmetric in F and G .

- 5) $I(P; F, G) \geq m_p(F) \cdot m_p(G)$, with equality if and only if F and G share no common tangents at P . If $I(P; F, G) = 1$, we say that the intersection of F and G is transversal at P .
- 6) $I(P; F \cdot \tilde{F}, G) = I(P; F, G) + I(P; \tilde{F}, G)$
- 7) If $A \in K[X, Y]$ and $G + AF$, then

$$I(P; F, G) = I(P; F, G + AF).$$

As before, a generalization to projective curves follows by considering affine representatives. Indeed, if P is a point in $\mathbb{P}^2$, choose coordinates such that P does not lie on $V(Z)$. Then, we dehomogenize the forms F and G at the coordinate Z (that is, we set $Z = 1$) and denote their dehomogenizations by F_* and G_* , respectively. The intersection number of F and G at P is defined to be $I(P_*; G_*, F_*)$ where P_* denotes the affine point corresponding to P . The properties from the above definition all carry over to the projective case. Noting that $\dim_k(\mathcal{O}_p(\mathbb{A}^2)/(F_*, G_*)) = \dim_k(\mathcal{O}_p(\mathbb{P}^2)/(F, G))$ holds, we obtain a similar formula for the intersection number of forms as in Definition 2.2.6.

The following example illustrates this definition. Consider the two projective curves $V(Y)$, $V(YZ - X^2)$ and the point $P = [0:0:1]$, which are sketched in the following figure.

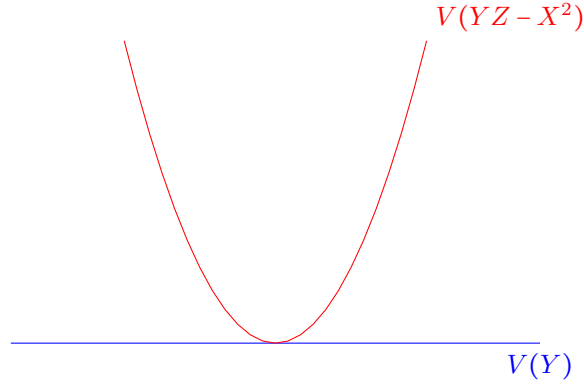


Figure 2: The curves $V(Y)$ and $V(YZ - X^2)$.

The two curves intersect at $P = [0:0:1]$, so the intersection number $I(P; Y, YZ - X^2)$ is nonnegative. The figure also convinces us that no matter how slightly we perturb the line $V(Y)$, we get two intersection points (perturbing downwards yields complex points in the above figure). Heuristically, we thus expect $I(P; Y, YZ - X^2) = 2$ to hold. Indeed, dehomogenizing both forms at Z , we obtain polynomials $F = Y$ and $G = Y - X^2$. By property 7) of Definition 2.2.6 we obtain $I(P; Y, YZ - X^2) = I(P; Y, X^2)$. Using properties 6) and 5), we then obtain $I(P; Y, X^2) = 2I(P; Y, X) = 2$ as desired.

An important result about intersection numbers is Bézout's theorem.

Theorem 2.2.7 (Bézout’s Theorem). *Let F and G be two coprime forms of degrees m and n , respectively. Then the equation*

$$m \cdot n = \sum_{P \in \mathbb{P}^2} I(P; F, G)$$

holds. In particular, the above sum is finite.

A proof of Bézout’s theorem may be found in [6].

Using Bézout’s theorem, one may prove the second main tool we need from intersection theory: Noether’s $AF + BG$ lemma. In essence, Noether’s $AF + BG$ lemma tells us that knowing local information at every point is enough to determine global information. Noether’s lemma is key in our proof of the Riemann-Roch theorem in Section 8.

Definition 2.2.8 (Noether’s conditions). Suppose F, G and H are three forms in $k[X, Y, Z]$ of degrees m, n and k respectively and suppose $P \in \mathbb{P}^2$. Choose a form L of degree one such that P does not lie in $V(L)$ and consider the elements in $\mathcal{O}_P(\mathbb{P}^2)$ given by

$$F_* := \frac{F}{L^m}, G_* := \frac{G}{L^n}, H_* := \frac{H}{L^k}.$$

We say that Noether’s Conditions are satisfied for F, G and H at P if H_* lies in the ideal generated by F_* and G_* in $\mathcal{O}_P(\mathbb{P}^2)$.

We attempt to give a geometric interpretation of these conditions. The conditions assert that $(F_*, G_*) = (F_*, G_*, H_*)$, thus the ideals (F_*, H_*) and (G_*, H_*) are both contained in the ideal (F_*, G_*) . In particular, we have

$$\begin{aligned} \dim_k(\mathcal{O}_P(\mathbb{P}^2)/(F_*, G_*)) &\leq \dim_k(\mathcal{O}_P(\mathbb{P}^2)/(F_*, H_*)), \\ \dim_k(\mathcal{O}_P(\mathbb{P}^2)/(F_*, G_*)) &\leq \dim_k(\mathcal{O}_P(\mathbb{P}^2)/(G_*, H_*)). \end{aligned}$$

By Definition 2.2.6, this is equivalent to stating that

$$\begin{aligned} I(P; F, G) &\leq I(P; F, H), \\ I(P; F, G) &\leq I(P; G, H). \end{aligned}$$

These conditions are vacuously true when P does not lie on $V(F) \cap V(G)$. However, when P is a point on $V(F) \cap V(G)$, Noether’s conditions can be interpreted as stating that P also lies on H and that H intersects both F and G “at least as much” in P as G and H intersect each other at P .

Lemma 2.2.9 (Noether’s $AF + BG$ lemma). *Suppose F, G , and H are forms of degree m, n , and k , respectively. Furthermore, Suppose Noether’s conditions are satisfied for F, G , and H for every $P \in \mathbb{P}^2$. Then $k \geq m$, $k \geq n$, and there exist forms A and B of degree $k - m$ and $k - n$, such that $H = AF + BG$.*

For a proof of this lemma, see [5].

2.3 Rational Maps

We end this section by recalling a few facts about rational maps and birational morphisms. First, we recall the definition of a rational map.

Definition 2.3.1. Let X and Y be varieties. Consider the space R of all pairs (U, f) where U is a nonempty open of X and $f: U \rightarrow Y$ is a morphism. We call pairs (U, f) and (V, g) equivalent if f and g agree on $U \cap V$. This relation defines an equivalence relation $\sim$ on R . A rational map $\Phi: X \dashrightarrow Y$ is an element of $R/\sim$, and any element $(U, f) \in \Phi$ is said to represent Φ . The domain of Φ is defined as the union of those opens which occur in the equivalence class. A rational map $\Phi: X \dashrightarrow Y$ is called dominating if the image of one of its representatives (and hence all) is dense in Y .

One may verify that the set of varieties equipped with rational maps forms a category. Since every morphism determines a unique rational map, the category of varieties with morphisms admits an essentially surjective and faithful embedding in this new category; it is not, however, fully faithful, as there are rational maps that do not come from global morphisms. In this new category, the right notion of isomorphism is birational equivalence.

Definition 2.3.2. Let $\Phi: X \dashrightarrow Y$ be a rational map between varieties. Φ is called birational if it is represented by an isomorphism $f: U \rightarrow V \subseteq Y$. Two varieties are called birationally equivalent if a birational map exists between them.

The above definition guarantees that two varieties are birationally equivalent if and only if they have isomorphic function fields. This fact is proven in [5].

Corollary 2.3.2.1. *Every algebraic curve is birationally equivalent to an affine plane curve.*

Proof. By Lemma 2.2.1, the function field of an algebraic curve may be written in the form $k(x, y)$. The affine curve with coordinate ring $k[x, y]$ has the same function field. $\square$

We will freely use several basic facts about morphisms and rational maps, summarized in the following lemma. Proofs are found in [5].

Lemma 2.3.3 (A few facts about morphisms and rational maps). *Let $f: C \dashrightarrow C'$ be a rational map between algebraic curves. Then the following statements hold*

- *If C is smooth and C' is projective, then f is a morphism.*
- *If f is a morphism and C is projective, then f is either constant or surjective.*
- *If f is a surjective morphism, then the induced map $f^\flat$ on function fields is injective.*
- *If f is birational and C and C' are projective, then f represents an isomorphism.*

3 Cremona Transformations

The following two sections aim to find smooth representatives of a given projective curve C . We would like to find a new smooth curve X and a birational morphism $f: X \rightarrow C$. This new smooth curve, referred to as the nonsingular model of C , will be unique up to unique isomorphism. Thus, when we study questions about C 's function field in Section 5, we will be able to restrict our attention to smooth curves without losing any birational information. Throughout most of this section, we follow [5]. However, our discussion of positive characteristic has extended the arguments found there.

3.1 The Standard Cremona Transformation

Our starting point in this section is a projective plane curve C , which may have nonordinary singularities (recall Definition 2.2.5). We will birationally transform C to remove such behavior, forcing C to have only ordinary singularities. In the next section, we will see how to transform a curve with only ordinary singularities into a smooth curve, which will help us complete our desingularization.

The main result of this section lets us pass from an arbitrary algebraic curve to one with only ordinary singularities. The theorem states

Theorem 3.1.1. *Every algebraic curve is birationally equivalent to a planar projective curve with only ordinary singularities.*

To achieve ordinary singularities on the curve C , we will primarily use a transformation known as the standard Cremona transformation. As we go through this section, let's refer to the points $[1 : 0 : 0]$, $[0 : 1 : 0]$, and $[0 : 0 : 1]$ as P_1, P_2 , and P_3 respectively, as these points will have special significance in the upcoming discussion.

Definition 3.1.2. Let U be the open subset of $\mathbb{P}^2$ given by $\mathbb{P}^2 \setminus \{P_1, P_2, P_3\}$. The standard quadratic transformation or standard Cremona transformation of $\mathbb{P}^2$ is given by the map

$$Q: U \rightarrow \mathbb{P}^2, [x : y : z] \mapsto [yz : xz : xy].$$

Remark. The standard Cremona map is a birational equivalence of the plane. Indeed, if we exclude the lines $V(X)$, $V(Y)$, and $V(Z)$ from the domain, it even becomes an involution, as the following calculation shows.

$$Q \circ Q([x : y : z]) = Q([yz : xz : xy]) = [x^2yz : y^2xz : z^2xy] = [x : y : z].$$

Given appropriate coordinates of $\mathbb{P}^2$, we can use the standard Cremona transformation to remove nonordinary singularities. We need to be careful about how C intersects the coordinate axes in order for this to work. Assuming the notation introduced in Definition 3.1.2, we define the following.

Definition 3.1.3. A curve $C \subseteq \mathbb{P}^2$ is called well-placed if the following conditions are met.

- The points P_1 and P_2 do not lie on C .
- Neither of the lines $V(X)$ nor $V(Y)$ is tangent to C at P_3 .
- If P_3 has multiplicity m_{P_3} on C , then $V(X)$ and $V(Y)$ each intersect C transversally in $n - m_{P_3}$ pairwise distinct points other than P_3 , where n is the degree of the (irreducible) form defining C .
- The line $V(Z)$ intersects C transversely in n distinct points.

Given a nonordinary singular point $P \in C$, we would like to change coordinates so that C becomes well-placed and this problematic point becomes P_3 . In $\text{char}(k) = 0$, we will prove that this is always possible; however, in $\text{char}(k) = p > 0$, P can be a so-called terrible point on C (we will define this later), in which case we can not find appropriate coordinates. We will show that in this case, we may first transform C birationally so that P is no longer terrible, after which we can then make C well-placed.

Provided that C is well-placed, we obtain the following theorem.

Theorem 3.1.4. Let $C = V(F)$ be a well-placed curve where F is an irreducible form of degree n . Let P_3 have multiplicity m_{P_3} on C and $C' = V(F')$ be the closure of the curve

$$Q^{-1}(C \setminus V(XYZ)) = Q(C \setminus V(XYZ)),$$

in $\mathbb{P}^2$. Then, $C \setminus V(XYZ)$ is isomorphic to $C' \setminus V(XYZ)$. Furthermore, the points P_1 , P_2 and P_3 are ordinary on C' with corresponding multiplicities $n - m_{P_3}$, $n - m_{P_3}$ and n . The curve C' does not intersect $V(XY)$ in any other points. Lastly, if C' intersects $V(Z)$ in further points $Q_1, \dots, Q_s$ then the equation $m_{P_3} = \sum_{i=1}^s I(Q_i, F' \cap Z)$ holds.

Remark. Since the theorem states that $C \setminus V(XYZ)$ is isomorphic to $C' \setminus V(XYZ)$, singular points on these open sets are in one-to-one correspondence. This correspondence preserves multiplicity and ordinarity.

Proof. First, write F as

$$F_r(X, Y)Z^{n-r} + \dots + F_{n-1}(X, Y)Z + F_n(X, Y). \quad (2)$$

where F_i is a form in X, Y of degree i and r is the multiplicity of C at P_3 . The points mapping to C under Q are given by the points satisfying the equation

$$\begin{aligned} 0 &= F(YZ, XZ, XY) \\ &= F_r(YZ, XZ)(XY)^{n-r} + \dots + F_{n-1}(YZ, XZ)(YX) + F_n(YZ, XZ) \\ &= Z^r (F_r(Y, X)(XY)^{n-r} + \dots + F_{n-1}(Y, X)Z^{n-1-r}YX + F_n(Y, X)Z^{n-r}) \end{aligned}$$

Since C' is a variety, it corresponds to an irreducible factor of the above equation; however, it does not correspond to Z^r since this line maps entirely to P_3 . We claim that the polynomial:

$$F' = F_r(Y, X)(XY)^{n-r} + \cdots + F_{n-1}(Y, X)Z^{n-1-r}YX + F_n(Y, X)Z^{n-r} \quad (3)$$

is irreducible, thus it is the form that defines C' . As stated, the points in $V(F(YZ, XZ, XY))$ are precisely those points that map to $V(F)$ under Q . Hence, neither X nor Y divides $F(YZ, XZ, XY)$ for these lines map entirely to $[1 : 0 : 0]$ and $[0 : 1 : 0]$ respectively, which by assumption, do not lie on C . We also know that F' is not divisible by Z since Z does not divide F_r . Now, we may write F' as

$$F'(X, Y, Z) = \frac{F(YZ, XZ, XY)}{Z^r}$$

and thus, since F is a form, we have the equation

$$\begin{aligned} F'(YZ, XZ, XY) &= (XYZ)^n \frac{F(X, Y, Z)}{(XY)^r} \\ &= (XY)^{n-r} Z^n F(X, Y, Z). \end{aligned}$$

If F' had a nontrivial factorization $F' = GH$ we would obtain

$$G(YZ, XZ, XY) \cdot H(YZ, XZ, XY) = (XY)^{n-r} Z^n F(X, Y, Z)$$

Furthermore since F' is not divisible by X , Y nor Z , neither is G nor H . In particular, $G(YZ, XZ, XY)$ and $H(YZ, XZ, XY)$ must each have a factor different from X , Y , or Z . Indeed, if these were the only factors of one of the two forms, the only points mapping to $V(G)$ or to $V(H)$ would be on these lines. We would then have $Q(Q^{-1}(V(G))) \subseteq \{P_1, P_2, P_3\}$, or $Q(Q^{-1}(V(H))) \subseteq \{P_1, P_2, P_3\}$ which is not possible unless the original forms correspond to some combination of the lines $V(X)$, $V(Y)$, and $V(Z)$. Hence, we obtain that F must have nontrivial factors, which is a contradiction.

It remains to show that the points P_1 , P_2 and P_3 are ordinary on $C' = V(F')$. Let us first calculate their multiplicities. This is done easiest for P_3 , since from Equation (3) we see that the lowest term of F' , its expansion about P_3 , is $F_n(Y, X)Z^{n-r}$ (which can be seen by dehomogenizing at Z). Thus $m_{P_3} = \deg(F_n(Y, X)) = n$. To show that P_3 is ordinary on F' , note that by Equation (2), the zeros of F_n correspond to the intersection points of C with $V(Z)$. Since we assumed C to be well-placed, these are n distinct points. In particular, the lowest term of F' expanded about P_3 has distinct tangents, one for every intersection point of C with $V(Z)$. This shows that P_3 is ordinary on C' .

Next, we expand F around $P_1 = [1 : 0 : 0]$ as

$$F = \tilde{F}_0(Y, Z)X^n + \cdots + \tilde{F}_n(Y, Z)X^0$$

Where $\tilde{F}_i$ is a form in $k[Y, Z]$ of degree i and $\tilde{F}_0$ is nonzero since C is well-placed. Then we may also write F' as

$$F' = Z^{-r}(\tilde{F}_0(XZ, XY)(YZ)^n + \cdots + \tilde{F}_n(XZ, XY)).$$

As before, $\tilde{F}_n(Z, Y)Z^{-r}X^n$ is the lowest term of the expansion of F' about P_1 ; therefore, $m_{P_1} = \deg(\tilde{F}_n(Z, Y)Z^{-r}) = n - r$. Now $\tilde{F}_n$ factors into exactly n lines where each line corresponds to an intersection point of C with $V(X)$. Since C is well-placed, we may write

$$\tilde{F}_n(Y, Z) = Y^r \prod_{i=1}^{n-r} L_i(Y, Z),$$

where the L_i are pairwise distinct. Here, we used that the factor Y corresponds to P_3 , and the other factors correspond to the other $n - r$ distinct intersection points of C and $V(X)$. In particular, we then have

$$\tilde{F}_n(Z, Y)Z^{-r} = \prod_{i=1}^{n-r} L_i(Y, Z).$$

Thus, the tangents of C' at P_1 are all distinct, which means that P_1 is ordinary on C' . Analogously, we may show that P_2 is ordinary on C' and also has multiplicity $n - r$. Finally, from Equation (3), we see that F' is a form of degree $2n - r$. Hence, taking into account the above degrees and multiplicities, Bézout's theorem implies that C does not intersect $V(X)$ or $V(Y)$ in any additional points, and C must intersect $V(Z)$ in additional points $Q_1, \dots, Q_s$ (when $r > 0$) such that $r = \sum_{i=1}^s I(P; C, Z)$. $\square$

3.2 Well-Placed Curves in Positive Characteristic

The main work required to prove the central result in this section is contained in the proof of Theorem 3.1.4. Before we prove Theorem 3.1.1, we will briefly discuss the subtleties that can arise when $\text{char}(k) = p > 0$.

When the base field has positive characteristic, the curve may contain particularly bad singularities, which is captured in the following definition taken from Appendix A of [5].

Definition 3.2.1. Let $F \in k[X, Y, X]$ be an irreducible form of degree n and $C := V(F)$. A point $P \in \mathbb{P}^2$ with multiplicity $r \geq 0$ on C is called terrible for C or terrible for F if an infinite number of lines through P intersect C in fewer than $n - r$ points other than P (not counting multiplicity of intersections).

Remark. The Definition 3.2.1 does not require that the terrible point lies on C since we allowed the case when $m_P(F) = 0$.

Remark. By Bézout's theorem, any line through P intersects F in $n - r$ points other than P when counted with multiplicity, and thus being terrible means that at one of these intersection points, the intersection must always occur in higher multiplicity.

A point P being terrible is equivalent to requiring that an infinite number of lines through P are tangent to C .

Terrible points do occur. Consider the curve defined over a field of characteristic $p > 0$ as $C := V(F)$ where $F := X^{p+1} - Y^p Z$. The form F is irreducible as $-Y^p Z$ is not a $(p+1)^{\text{th}}$ power in the field $k(y, z)$ (here we use Gauss's lemma to ensure irreducibility). Furthermore, the point $P = [0 : 1 : 0]$ lies on C and has multiplicity $r = 1$, which can be seen by dehomogenizing at Y . Lines through P are given by sets of the form $V(aX + bZ)$ where $a, b \in k$ are not both zero and the set of lines of the form $V(X - aZ)$ forms an infinite subset of such lines. If Q is an intersection point of $V(aX - Z)$ and C , which is not P , it has the form $Q = [a : y : 1]$. Solving $F([a : y : 1]) = 0$ for y we obtain the equation

$$y^p - a^{p+1} = 0,$$

which has a unique solution, since the polynomial $T^p - a^{p+1}$ has vanishing derivative. In particular, all these lines intersect C in only one point other than P instead of $p = (p+1) - 1$ points.

In the next lemma, we will find a necessary condition on the form F to have a terrible point. Our strategy to remove the terrible point from F is to birationally change the curve $C = V(F)$ such that the form defining the transformed curve can no longer fulfill this condition.

Lemma 3.2.2. *Suppose $C = V(F)$ is a projective plane curve and P is a terrible point for C of multiplicity r on C . If n is the degree of F then p divides $n - r$.*

Proof. Without loss of generality, let $P = [0 : 0 : 1]$ since projective transformations leave degrees and multiplicities unchanged. The lines through P are all of the form $V(\alpha X + \beta Y)$. When $\alpha \neq 0$ holds, these lines can be written as $V(X - aY)$. Then, the intersection points that are different from P , are of the form $[a : 1 : \lambda]$, which means they are determined uniquely by the equation (in λ):

$$G_a(\lambda) := F(a, 1, \lambda) = 0.$$

The assumption states that for infinitely many values $a \in k$, the polynomial G_a has a double root. If we expand F around P we may represent F as

$$F(X, Y, Z) = Z^{n-r} F_r(X, Y) + Z^{n-r-1} F_{r+1}(X, Y) + \cdots + F_n(X, Y)$$

where F_i is either a form of degree i in $k[X, Y]$ or zero and $F_r \neq 0$. Then, we may write G as

$$G(\lambda) = \lambda^{n-r} F_r(a, 1) + \lambda^{n-r-1} F_{r+1}(a, 1) + \cdots + F_n(a, 1).$$

If we regard F as a polynomial in Z over the field $k(x, y)$, then F is irreducible and has $n - r$ distinct roots (in some algebraic closure) unless p divides $n - r$, since otherwise F and F_Z are coprime. If p does not divide $n - r$, then G has a double root if and only if it has a common factor with its derivative in λ , which in turn defines a common intersection point of F and F_Z . In this case, since F_Z is not zero, there can only be finitely many such intersection points. $\square$

In the following, we want to show that there can be at most one point P , which is terrible for F . To do this, we will need to consider the dual curve. The dual curve is closely related to the usual notion of duality of the projective plane between lines and points. Roughly speaking, the dual curve of an algebraic curve will be a variety consisting of tangent lines to the variety.

Theorem 3.2.3 (The dual curve). *Suppose C is a projective planar curve given by an irreducible form F of degree $n > 1$ with partial derivatives F_X, F_Y, F_Z . Then there exists a unique plane curve C' , called the dual curve to C , which contains all the points of the form $[F_X(P) : F_Y(P) : F_Z(P)]$ for simple points $P \in C$.*

Proof. First, note that not all the partial derivatives F_X, F_Y, F_Z can be zero, or else F would be reducible (a precise argument was seen in Section 2). In particular $V(F_X, F_Y, F_Z)$ is a proper closed subset of $\mathbb{P}^2$ and the open $U := \mathbb{P}^2 \setminus V(F_X, F_Y, F_Z)$ is not the empty set. We define the rational map Φ represented by the morphism:

$$\Phi: U \rightarrow \mathbb{P}^2, P \mapsto [F_X(P) : F_Y(P) : F_Z(P)].$$

Define the open on C by $U' := C \cap U$. Since not all partials are zero, Bézout's theorem ensures that this set is not the empty set either, and Φ restricted to U defines a dominating rational map:

$$\Phi: C \dashrightarrow \overline{\Phi(U')}, P \mapsto [F_X(P) : F_Y(P) : F_Z(P)].$$

We claim that $C' := \overline{\Phi(U')}$ is the desired curve. First, $\overline{\Phi(U')}$ is a variety since $\Phi(U')$ is the image of an irreducible. In particular, $\overline{\Phi(U')}$ is either a point or another curve. If $\overline{\Phi(U')}$ were a single point say $\{[a : b : c]\}$ then for every point P in U' on C we would have:

$$[F_X(P) : F_Y(P) : F_Z(P)] = [a : b : c],$$

thus the line $L = V(aX + bY + cZ)$ would intersect F in every simple point of C . Since the latter set is infinite, Bézout's theorem implies that L and C must have a common component, which is impossible since F has a degree strictly larger than one and is irreducible. The curve C' is unique, as any other plane curve containing all these points intersects C' infinitely often; thus, since C' is a variety, it must contain C' as a component. $\square$

Remark. If F defines a line, the transformation Φ in the above proof sends $V(F)$ to its dual point, so Theorem 3.2.3 is a generalization of projective duality.

Corollary 3.2.3.1. *Let $C = V(F)$ be a projective plane curve. Then there is at most one point P , which is terrible for F .*

Proof. Suppose P is a terrible point for F of multiplicity r . Then there are an infinite number of lines through P which intersect C in fewer than $n - r$ distinct

points other than P . By Remark 2.2, C has only finitely many singularities. Thus an infinite number of these lines are, in fact, tangents to C at smooth points. If we write these tangents as $V(aX + bY + cZ)$, then by Theorem 3.2.3, the dual curve of F contains all the points of the form $[a : b : c]$. Writing $P = [p_1 : p_2 : p_3]$, all these points lie on the line $L = V(p_1X + p_2Y + p_3Z)$. It follows that the dual contains an infinite number of points on this line. Since the dual is closed and irreducible, it must be equal to this line, so the dual is uniquely determined by the terrible point, from which it follows that there can be only one terrible point. $\square$

We can now prove an important intermediate result, which states that we may choose coordinates such that the $V(Z)$ in $\mathbb{P}^2$ intersects C transversely in n distinct points.

Corollary 3.2.3.2. *Let $C \subseteq \mathbb{P}^2$ be an algebraic curve defined by an irreducible form F of degree n . Then, there exists a line that intersects C in n distinct points.*

Proof. By Corollary 3.2.3.1, there is at most one point P , which is terrible for C . Let P be any non-terrible point for C , which does not lie on C . Then all but finitely many lines through P intersect C in n distinct points. $\square$

Remark. The above proof shows that for any non-terrible point for C which does not lie on C , all but finitely many lines through this point intersect C transversely.

3.3 The Projective Space of Forms

As a final preparation for the main result in this section (Theorem 3.1.1), we will need one more technical result. The proof of 3.1.1 is roughly outlined as follows. To each plane curve C , we will assign a unique nonnegative integer called $g^*(C)$. Successively, we will birationally transform C to obtain a sequence of plane curves

$$C = C_1 \rightarrow C_2 \rightarrow \cdots \rightarrow C_n.$$

In each step, we show that if we do not yet have a curve of the desired type, it is possible to choose another transformation that either decreases the total number of nonordinary singularities or decreases the integer $g^*(C_i)$ and such that it never increases $g^*(C_i)$. Since both of these quantities are nonnegative integers, this process cannot continue indefinitely. Thus, we will eventually obtain a curve with no nonordinary singularities. The manner in which we will define $g^*(C)$ does not immediately assure us of its nonnegativity; thus, we will first prove a few useful results for this.

Fix an integer $d \geq 0$ and let $W(d)$ be the vector space of forms in $k[X, Y, Z]$, which have degree d . Clearly the monomials $\{M_{i,j,k} = X^i Y^j Z^k \mid i + j + k = d\}$

form a k -basis of $W(d)$. This allows us to calculate the dimension of $W(d)$ as:

$$\dim_k(W(d)) = \sum_{i=0}^d \sum_{j=0}^{d-i} 1 = \sum_{i=0}^d (d-i+1) = \binom{d+2}{2}.$$

If we order the $M_{i,j,k}$ in some manner $M_1, \dots, M_t$ where $t = \binom{d+2}{2}$. Then, every form of degree d uniquely determines a nonzero tuple of coefficients $(\lambda_1, \dots, \lambda_t)$. Two such forms determine the same algebraic set if and only if they differ in some nonzero scalar. Thus, if we allow curves with multiple components (possibly even of multiplicity), the space V_d of curves defined by forms of degree d may be identified with a projective space of projective dimension $\binom{d+2}{2} - 1 = \frac{d(d+3)}{2}$.

One may observe that if $T: \mathbb{P}^2 \rightarrow \mathbb{P}^2$ is some projective change of coordinates, then the map $F \rightarrow F^T = F \circ T$ defines a unique projective change of coordinates of V_m , which is important in the following lemma.

Lemma 3.3.1. *Let $P_1, \dots, P_n$ be points in $\mathbb{P}^2$ and let $d, r_1, \dots, r_n$ be positive integers. Furthermore, denote by $V(d, r_1 P_1, \dots, r_n P_n)$ the subset of V_d consisting of the forms that pass through each P_i with multiplicity at least r_i . Then $V(d, r_1 P_1, \dots, r_n P_n)$ is a projective subspace of V_d , and its projective dimension is at least $\frac{d(d+3)}{2} - \sum_{i=1}^n \frac{r_i(r_i+1)}{2}$.*

Proof. First, consider the case when $n = 1$ and $r_1 = 1$. Then, we are asking for the projective space of forms that satisfy $F(P) = 0$. Choosing some change of coordinates, we may assume $P = [0 : 0 : 1]$ since this will induce a projective change of coordinates in V_d . We may expand a form of degree d about P as:

$$F(X, Y, Z) = \lambda Z^d + F_1(X, Y)Z^{d-1} + \dots + F_k(X, Y)Z^{d-k}.$$

Where F_i is a form of degree i and $F_k \neq 0$. From this, we see that $F(P) = 0$ occurs if and only if Z^d appears with coefficient zero. Thus, the set of forms that vanish at P form a hyperplane in V_d . Now we consider the case $r_1 > 1$; we may also use the same argument as above by writing

$$F(X, Y, Z) = \lambda Z^d + F_1(X, Y)Z^{d-1} + \dots + F_k(X, Y)Z^{d-k},$$

and recognizing that in this representation the multiplicity of F at P is at least r_1 if and only if $\lambda = F_1 = \dots = F_{r_1-1} = 0$. Hence, the space $V(d, r_1 P_1)$ forms a linear projective subspace of V_d , and we are left to calculate its dimension. The restriction on $V(d, r_1 P_1)$ implies that $Z^i F_{d-i}(X, Y) = 0$ for $i = d, \dots, d - r_1 + 1$, which is equivalent to stating that the coefficients of $Z^i X^j Y^k$ are zero whenever $i + j + k = d$ and $j + k < r_1$ hold. Thus, the problem reduces to counting how many such triples there are. This is calculated as:

$$\sum_{s=0}^{r_1-1} |\{(j, k) \mid j + k = s\}| = \sum_{s=0}^{r_1-1} (s+1) = \frac{r_1(r_1+1)}{2},$$

which implies that the projective dimension of $V(d, r_1 P_1)$ is given by

$$\frac{d(d+3)}{2} - \frac{r_1(r_1+1)}{2}.$$

Now suppose $n > 1$. We may write:

$$V(d, r_1 P_1, \dots, r_n P_n) = \bigcap_{i=1}^n V(d, r_i P_i)$$

Which is an intersection of projective subspaces and thus has a dimension at least

$$\frac{d(d+3)}{2} - \sum_{i=1}^n \frac{r_i(r_i+1)}{2}.$$

□

Corollary 3.3.1.1. *Let $C = V(F)$ be a projective plane curve of degree $n := \deg(F)$. Let $P_1, \dots, P_m$ be the multiple points of C with multiplicities $r_1, \dots, r_m$ respectively. Then, the quantity*

$$g^*(C) := \frac{(n-1)(n-2)}{2} - \sum_{i=1}^m \frac{r_i(r_i-1)}{2},$$

is nonnegative.

Proof. We first claim that the quantity

$$K := \frac{(n-1)n}{2} - \sum_{i=1}^m \frac{r_i(r_i-1)}{2},$$

is nonnegative. Indeed, we may choose coordinates such that $F_X \neq 0$. Then F_X is of degree $n-1$, and at every point $P \in V(F)$, the form F_X has multiplicity at least $r_P - 1$. If we apply Bézout's theorem and property 5) from 2.2.6 to F and F_X , we obtain

$$\begin{aligned} n(n-1) &= \sum_{P \in C} I(P; F, F_X) \\ &\geq \sum_{P \in C} m_P(F) m_P(F_X) \\ &\geq \sum_{P \in C} m_P(F) (m_P(F) - 1) \\ &\geq \sum_{i=1}^m r_i(r_i - 1). \end{aligned}$$

Which shows $K \geq 0$. Observing

$$\alpha := \frac{(n-1)(n+2)}{2} - \sum_{i=1}^m \frac{r_i(r_i-1)}{2} \geq K/2 \geq 0,$$

and $\alpha \in \mathbb{Z}$, we may proceed as follows. Since C has finitely many singular points, we may choose α smooth points on C say $\tilde{P}_1, \dots, \tilde{P}_\alpha$. Then we have

$$\frac{(n-1)(n+2)}{2} - \sum_{i=1}^m \frac{r_i(r_i-1)}{2} - \sum_{i=1}^\alpha \frac{2}{2} = \alpha - \alpha = 0.$$

Lemma 3.3.1 then guarantees the existence of a form G of degree $n - 1$ with $m_{P_i}(G) \geq r_i - 1$ for $i = 1, \dots, m$ and $m_{\tilde{P}_j}(G) \geq 1$ for $j = 1, \dots, \alpha$. If we apply Bézout's theorem and property 5) from Lemma 2.2.6 once again, we obtain

$$\begin{aligned}
n(n-1) &= \sum_{P \in C} I(P; F, G) \\
&\geq \sum_{P \in C} m_P(F) m_P(G) \\
&\geq \sum_{i=1}^m r_i(r_i - 1) + \sum_{j=1}^{\alpha} 1 \\
&= \sum_{i=1}^m r_i(r_i - 1) + \frac{(n-1)(n-2)}{2} - \sum_{i=1}^m \frac{r_i(r_i - 1)}{2}.
\end{aligned}$$

Rearranging the above inequality, this becomes

$$\frac{(n-1)(n-2)}{2} - \sum_{P \in C} \frac{r_P(r_P - 1)}{2} \geq 0$$

which finishes the proof. $\square$

We are now ready to prove the main result of this section, and in doing so, we will complete the first step in proving the existence of nonsingular models.

(*Proof of Theorem 3.1.1*). Let C be an algebraic curve with function field K . Since K is a field of dimension one over the algebraically closed field k , we may choose $x, y \in K$ such that $K = k(x, y)$ by Lemma 2.2.1. The affine curve with coordinate ring $k[x, y]$ is birationally equivalent to C and is planar; thus, its projective completion is projective planar. Let F be the irreducible form defining the plane curve $C \subseteq \mathbb{P}^2$ and n its degree.

The main idea of the proof is to successfully apply standard quadratic transformations to C , eventually making all singularities ordinary. To ensure this process eventually stops, consider the quantity:

$$g^*(C) = \frac{(n-1)(n-2)}{2} - \sum_{P \in C} \frac{r_P(r_P - 1)}{2}.$$

Where r_P denotes the multiplicity of C at P and n is the degree of C . As we have seen in Corollary 3.3.1.1 $g^*(C)$ is always a nonnegative integer. We will now successively transform C such that in each step, we either decrease the number of nonordinary singularities or decrease the quantity g^* . Since both quantities are nonnegative integers, this process must eventually terminate.

Let Q be any nonordinary point of C , and let its multiplicity be $r > 0$. First, consider the case when Q is terrible for F . Not both n and $n - 1$ can be divisible by p so choose a point $P \neq Q$ of multiplicity $m \in \{0, 1\}$ such that $n - m$ is not divisible by p . Furthermore, take coordinates such that $P = [0 : 0 : 1]$ and such that C is well-placed with Q not lying on $V(XYZ)$. This is possible since P is not terrible and by the remark after Corollary 3.2.3.2. Now perform

the standard quadratic transformation from Definition 3.1.2 at P . The proof of Theorem 3.1.4 shows that the transformed curve has degree $2n - m$ and the image of Q still has multiplicity r . By Lemma 3.2.2 p divides $n - r$. Therefore, p does not divide $2n - m - r$ by choice of m , which implies that the image of Q is no longer terrible for F .

If C' is the new curve, then by Theorem 3.1.4, the above process introduced no nonordinary singularities (note here that we require the fact $m \in \{0, 1\}$ so that the new points on $V(Z)$ are simple). In particular we may calculate $g^*(C')$ as

$$g^*(C') = \frac{(2n - m - 1)(2n - m - 2)}{2} - \sum_{P \in C'} \frac{r_P(r_P - 1)}{2}$$

Using Theorem 3.1.4, we can rewrite this as

$$\begin{aligned} g^*(C') &= \frac{(2n - m - 1)(2n - m - 2)}{2} - \sum_{P \in C} \frac{r_P(r_P - 1)}{2} - 2 \frac{(n - m)(n - m - 1)}{2} - \frac{n(n - 1)}{2} \\ &= \frac{n^2 - m^2 + m - 3n + 2}{2} - \sum_{P \in C} \frac{r_P(r_P - 1)}{2} \\ &= g^*(C) - \frac{m(m - 1)}{2} \\ &= g^*(C). \end{aligned}$$

Where we used the fact that $m \in \{0, 1\}$ for the last equation.

Based on the above discussion, we can now assume Q is not terrible on C , and thus, we may change coordinates such that $Q = [0 : 0 : 1]$ and that C is well-placed. We now perform a quadratic transformation at Q . Let C' be the transformed curve, if $P_1, \dots, P_S$ are the new points on $C' \cap V(Z)$ occurring in Theorem 3.1.4 we may calculate $g^*(C')$ as

$$\begin{aligned} g^*(C') &= \frac{(2n - r - 1)(2n - r - 2)}{2} - \sum_{P \in C'} \frac{r_P(r_P - 1)}{2} \\ &= \frac{(2n - r - 1)(2n - r - 2)}{2} - \left(\sum_{P \in C, P \neq Q} \frac{r_P(r_P - 1)}{2} \right) \\ &\quad - 2 \frac{(n - r)(n - r - 1)}{2} - \frac{n(n - 1)}{2} - \sum_{i=1}^s \frac{r_{P_i}(r_{P_i} - 1)}{2} \\ &= \left(\frac{n^2 - 3n + 2}{2} - \frac{r(r - 1)}{2} - \sum_{P \in C, P \neq Q} \frac{r_P(r_P - 1)}{2} \right) - \sum_{i=1}^s \frac{r_{P_i}(r_{P_i} - 1)}{2} \\ &= g^*(C) - \sum_{i=1}^s \frac{r_{P_i}(r_{P_i} - 1)}{2} \leq g^*(C). \end{aligned}$$

Note that the only possible points where the transformation can introduce new nonordinary singularities are at one of the P_i and from the above equation. If this happens, the above equation reads $g^*(C') < g^*(C)$. If the transformation

did not create new nonordinary singularities, we have successfully decreased the number of nonordinary singularities. We repeat this process successively, choosing a remaining nonordinary singularity in each step and performing a Cremona transformation centered at it (if needed, we make this point non-terrible first). In each step, we either strictly decrease the quantity g^* or strictly reduce the total number of nonordinary singularities, and we never increase g^* . Since $g^*(C)$ is a finite number, the former case can only occur finitely often. Thus, eventually, we must arrive at a curve with only ordinary singularities. $\square$

For technical reasons, in later proofs, we will need to birationally transform our plane curve so that the degree of its defining form becomes large. We would like to do this without creating nonordinary singularities. The Cremona transformations discussed here can achieve this feature if we center them at the right points.

Lemma 3.3.2. *Let $C = V(F)$ be a plane curve with no nonordinary singularities with $\deg(F) = n$. Then C is birationally equivalent to a curve $C' = V(F')$, which also does not have any nonordinary singularities and for which $\deg(F') = 2n$ (as usual, we assume that F and F' are irreducible).*

Proof. Let P be any point not on C , which is not terrible for C . Then, we may choose coordinates so that $P = [0 : 0 : 1]$ and such that C is well-placed. We may perform a quadratic transformation at P . By Theorem 3.1.4, this does not create any new nonordinary singularities (note since P is not on C , we have $0 = \sum_{i=1}^s I(Q_i, F' \cap Z)$). By the proof of Theorem 3.1.4, we also see that $\deg(F') = 2n - 0 = 2n$. $\square$

4 The Blowup of a Curve

In the last section, we saw that we could birationally transform any algebraic curve into a plane curve with only ordinary singularities. If C is the curve in question and P is a point on C , we may be interested in evaluating a function $z = f/g$ in C 's function field at P . Yet it may happen that $g(P) = 0$, so we don't have a well-defined value. In the case $k = \mathbb{C}$, we have a solution for this problem. We may consider what limiting value $z(Q)$ approaches as we approach P by points $Q \in C$. If P is an ordinary singularity of the curve C , then C has distinct tangents at P , so there are different directions in which we can approach P in, on C . For each tangent line of C at P , we get a direction of approach to P on C , and for each such direction, the behavior of z may be very different. This is illustrated in the following diagram.

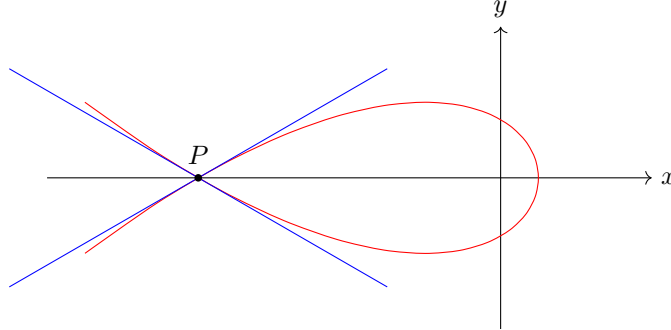


Figure 3: A curve with an ordinary double singularity

For arbitrary fields, this behavior is also seen; at such points, we have multiple local rings, each with maximal ideal consisting of a subset of the functions vanishing at P . In the following, we will construct a smooth curve X that admits a unique valuation at every point and which is birationally equivalent to C . The set of smooth points on C will be isomorphic to an open subset of X . For every multiple point on C , each tangent line will correspond to a unique point on this new curve. On this new smooth curve, there will be a one-to-one correspondence between discrete valuation rings in the function field and points on the curve. We can visualize the idea of the blowup in the figure above. Suppose the curve from the figure is parameterized as

$$\varphi: \mathbb{A}^1 \rightarrow \mathbb{A}^2, t \mapsto (\varphi_1(t), \varphi_2(t)).$$

Then, consider the curve parameterized by

$$\varphi': \mathbb{A}^1 \rightarrow \mathbb{A}^3, t \mapsto (\varphi_1(t), \varphi_2(t), t),$$

which is illustrated in Figure 4. Visually, we may convince ourselves that it is smooth and that the smooth points on the original curve lie under exactly one point. The ordinary singularity lies under two points since C “goes through it twice”. Strictly speaking, the blowup is, in general, not of this form; however, the general idea of passing to a higher dimension is the same. The main theorem of this section combines this idea with the work we have done in the previous section to obtain the existence of nonsingular models. It is stated as follows.

Theorem 4.0.1. *For any projective algebraic curve C , there exists a smooth projective curve X and a surjective birational morphism $f: X \rightarrow C$. Furthermore, the pair (X, f) is unique up to unique isomorphism; that is, if (Y, g) is another such pair for C , there exists a unique isomorphism $i: X \rightarrow Y$ such that $f = g \circ i$.*

Let us now begin our work to prove this theorem.

4.1 The Affine Blowup

We “blow up” the plane at a point P to construct the new curve. The idea is to replace the point P on the plane with an entire line. Each point on this line

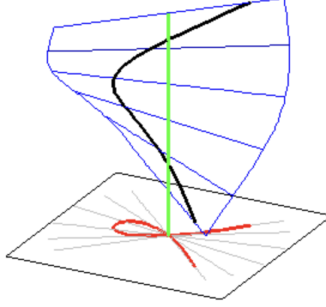


Figure 4: Blowup of a Curve with Ordinary Singularity [10]

will correspond to a unique direction in which we can approach P in, on the plane. For simplicity, we will first show this construction in the affine case in the following theorem.

Theorem 4.1.1. *Let $C \subseteq \mathbb{A}^2$ be a curve with an ordinary singularity at P of multiplicity r . Then, there exists a curve $C' \subseteq \mathbb{A}^2$ and a morphism $\Phi: C' \rightarrow C$ such that Φ maps exactly r smooth points of C' to P in the plane. Furthermore, if we exclude these r points in the domain of Φ , this map is an isomorphism with an open subset of C . In particular, these curves are birationally equivalent.*

Remark. The above theorem is sometimes also referred to as the “resolution of singularities”. A procedure such as the one we shall see in the following proof is also called the “resolving of singularities”.

Proof. Using a change of coordinates we may assume $P = (0,0)$, $C \neq V(X)$ and that X is not tangent to C at P . To show that the desired curve exists, we first construct a curve in $\mathbb{A}^3$, which satisfies the properties, then we find an isomorphic planar embedding of this curve. Consider the map given by

$$f: \mathbb{A}^2 \setminus V(X) \rightarrow \mathbb{A}^3, (x, y) \rightarrow (x, y, y/x).$$

The domain of this map is isomorphic to the affine variety $V := V(ZX - 1) \subseteq \mathbb{A}^3$ via the projection map $\pi_Z|_V (x, y, z) = (x, y)$. Now $f \circ (\pi_Z|_V)(x, y, z) = (x, y, yz)$ is a polynomial map between affine varieties and thus a morphism. Since $f = (f \circ (\pi_Z|_V)) \circ (\pi_Z|_V)^{-1}$, f is also a morphism. The image of f is given by the set

$$B = \{(x, y, z) \in \mathbb{A}^3 \mid y - xz, x \neq 0\},$$

which is an open subset of the variety $D = V(Y - ZX)$, in fact, D is the closure of B . Now restricting π_Z to D we get that D is mapped to the set

$(\mathbb{A}^2 \setminus V(X)) \cup \{P\}$ in such a way that $\pi_Z|_D$ is an isomorphism if we exclude those points above P and the preimage of P is given by $L = \{(0, 0, z) \mid z \in k\}$ which is closed in B . The map f will serve as the blowup map.

Next, we show that D is isomorphic to $\mathbb{A}^2$. Indeed, consider the map given by $\psi: \mathbb{A}^2 \rightarrow D, (x, z) \mapsto (x, xz, z)$. It is a morphism, as it is polynomial, and it is an isomorphism, as the projection $\pi_Y|_D$ is its inverse. These maps are visualized in the following commutative diagram where f is only a rational map and not a morphism.

$$\mathbb{A}^2 \begin{array}{c} \xrightarrow{\psi} \\ \xleftarrow{\pi_Y} \end{array} D \begin{array}{c} \xrightarrow{\pi_Z} \\ \xleftarrow[f]{-} \end{array} \mathbb{A}^2 \supseteq C.$$

If we define $\Phi := \pi_Z \circ \psi$, we get a birational morphism from the plane to itself, which will serve as a suitable transformation of C , which we will prove now. Let F be the irreducible form defining C . Since C has multiplicity r at $P = (0, 0)$, we may write F as

$$F(X, Y) = F_r(X, Y) + F_{r+1}(X, Y) + \cdots + F_n(X, Y),$$

where F_i is a form of degree i . Since $\Phi(x, z) = (x, xz)$ the points mapping to C all satisfy the equation

$$\begin{aligned} F(X, XZ) &= F_r(X, XZ) + F_{r+1}(X, XZ) + \cdots + F_n(X, XZ) \\ &= X^r (F_r(1, Z) + F_{r+1}(1, Z)X + \cdots + F_n(1, Z)X^{n-r}). \end{aligned}$$

The birational transform of C under Φ is a variety since Φ is a birational morphism, thus it corresponds to an irreducible factor of the polynomial in the above equation. It does not, however, correspond to X , since $V(X)$ maps entirely to $P = (0, 0)$. Hence it is an irreducible factor of

$$F' := F_r(1, Z) + F_{r+1}(1, Z)X + \cdots + F_n(1, Z)X^{n-r}.$$

We claim that F' is irreducible. Indeed, if we had some nontrivial factorization $F' = G(X, Z) \cdot H(X, Z)$, then we would also have:

$$F(X, Y) = X^r G(X, Y/X) \cdot H(X, Y/X).$$

Applying Gauss's Lemma to $k[x, y] \subseteq k[y](x)$, we get a factorization of F in $k[x, y]$, contradicting that C is a variety. Since ψ is an isomorphism and π_Z is an isomorphism between B and $\mathbb{A}^2 \setminus V(X)$, we get that Φ induces an isomorphism between $C' \setminus V(X)$ and $C \setminus V(X)$. Finally, the points on C' which map to P under Φ must have x -coordinate zero, and thus, they correspond to zeros in Z of $F_r(1, Z)$. Since P is ordinary on C with multiplicity r , we may factor $F_r(X, Y)$ into a product of r distinct lines $L_i = \alpha_i X + \beta_i Y$. This allows us to obtain

$$F_r(1, Z) = \prod_{i=1}^r (\alpha_i + \beta_i Z).$$

Since X is not tangent to C at P , none of the β_i are zero. Therefore, we obtain r roots given by $P_i = (0, -\alpha_i/\beta_i)$. Lastly, all tangents are distinct since P is ordinary on C . Thus the P_i are distinct, since $\alpha_i/\beta_i \neq \alpha_j/\beta_j$ for $i \neq j$. In summary, Φ maps $C' \setminus V(X)$ isomorphically to $C \setminus V(X)$ and the r distinct points of $C' \cap V(X)$ all map to P . From this, it must follow that the P_i are smooth on C . Indeed, by Lemma 2.2.6, we have the following inequality

$$m_{P_i}(C') \leq I(P_i; F', X) = I(P_i; F_r(1, Z), X) = 1.$$

□

4.2 The Projective Blowup

We will now try to generalize the idea from Theorem 4.1.1 to remove not one but many multiple points. Using the notation as in Theorem 4.1.1, one of the main issues of the affine blowup is that the map f is not defined on the points on $V(X)$ since these points do not have finite slope y/x . Furthermore, other than the origin, none of these points are in the image of Φ . Thus, in a way, these points are forgotten by the affine blowup. We would like our nonsingular model to project surjectively to points on C . If we allow infinite slopes y/x as well, we could extend the morphism f to the nonorigin points of $V(X)$; hence, changing to the projective setting is natural.

Theorem 4.2.1 (Projective blowup of a curve). *Let $C \subseteq \mathbb{P}^2$ be a projective curve with only ordinary singularities. Then, there exists a smooth projective curve X and a surjective morphism $\Phi: X \rightarrow C$ such that for any point P on C , exactly $m_P(C)$ points on X map to P . Furthermore, if we exclude from X the points mapping to singular points on C , the map Φ becomes an isomorphism with its image. In particular, these curves are birationally equivalent.*

Remark. As in the affine case, the procedure seen in the below proof is called the resolution of singularities.

Proof. Choose a line containing none of the finitely many multiple points of C , and change coordinates to make it the line at infinity ($V(Z)$). If $P_1, \dots, P_m$ are the multiple points of C , then in this coordinate system, they have coordinates $P_i = [a_i : b_i : 1]$. Let E be the set consisting of the P_i , and let U be the open set defined as $U := \mathbb{P}^2 \setminus E$. Define the m maps f_i by

$$f_i: U \rightarrow \mathbb{P}^1, [x : y : z] \mapsto [x - za_i : y - zb_i].$$

These maps are all well defined by choice of domain, and they are morphisms since restricting to any of the canonical affines of $\mathbb{P}^2$ and $\mathbb{P}^1$ yields polynomial maps. If we form the product of these maps, we get a morphism f given by

$$f: U \rightarrow \mathbb{P}^1 \times \dots \times \mathbb{P}^1, P \mapsto (f_1(P), \dots, f_m(P)).$$

This map will serve as the blowup of our original curve. Since U is a variety, the graph $G \subseteq U \times \mathbb{P}^1 \times \cdots \times \mathbb{P}^1$ of U under f is also a variety, and by definition, the points in G satisfy, the equations:

$$T_{i,1}(Y - b_i Z) - T_{i,2}(X - a_i Z), \text{ for } i = 1, \dots, m,$$

where X, Y, Z are homogeneous coordinates for $\mathbb{P}^2$ and $T_{i,1}, T_{i,2}$ are homogeneous coordinates for the i^{th} copy of $\mathbb{P}^1$ in the above product. In particular, since each of these equations is multi-homogeneous, the graph G is contained in the closed multi-homogeneous set D given by

$$D = V(T_{i,1}(Y - b_i Z) - T_{i,2}(X - a_i Z) \mid i = 1, \dots, m).$$

G is also an open subset of D since it is given by the complement of the union of the lines L_i defined below. As in the affine case, we will see that D is the closure of G in the space

$$\mathbb{P}^2 \times \mathbb{P}^1 \times \cdots \times \mathbb{P}^1.$$

Consider the projection $\pi: \mathbb{P}^2 \times \mathbb{P}^1 \times \cdots \times \mathbb{P}^1 \rightarrow \mathbb{P}^2$ mapping to the first coordinate, it is a morphism that maps surjectively to $\mathbb{P}^2$. By construction, π induces an isomorphism of G and U and thus is a birational equivalence from D to $\mathbb{P}^2$. If a point in D projects to the point P_i , then all of the coordinates $[T_{j,1} : T_{j,2}]$ for $j \neq i$ are uniquely determined by the relations defining D , and the i^{th} copy of $\mathbb{P}^1$ corresponding to the point P_i is free. So this set of such points is given by:

$$L_i := \{P_i\} \times \{f_1(P_i)\} \times \cdots \times \{f_{i-1}(P_i)\} \times \mathbb{P}^1 \times \{f_{i+1}(P_i)\} \times \cdots \times \{f_m(P_i)\}.$$

Clearly, L_i is isomorphic to a projective line.

To show that f transforms C in the desired way, we show that for a point $Q \in L_i$, the maps f and π locally look like the maps we considered in the affine blowup. Indeed, without loss of generality, let Q project to $[0 : 0 : 1]$, or else change the following maps in the same way as in Theorem 4.1.1. Furthermore, we assume Q is affine on L_i ; that is, we may write the $(i+1)^{\text{th}}$ coordinate of Q as $[\lambda : 1]$ for some $\lambda \in k$. Otherwise, we change coordinates in the i^{th} copy of $\mathbb{P}^1$. The same argument shows that we may also choose λ to be nonzero. Denote by $\varphi: \mathbb{A}^2 \rightarrow \mathbb{P}^2$ be the usual inclusion map mapping (x, y) to the projective point $[x : y : 1]$ and denote by $\psi: \mathbb{A}^2 \rightarrow \mathbb{A}^2$ the birational morphism we introduced in the affine blowup, mapping (x, z) to (x, xz) . Let $W \subseteq \mathbb{A}^2$ be the open set of points in $\mathbb{A}^2$, which are not mapped to any of the other P_j (for $j \neq i$) by the map $\varphi \circ \psi$. Finally, define the morphism $\Phi: W \rightarrow \mathbb{P}^2 \times \mathbb{P}^1 \times \cdots \times \mathbb{P}^1$ as

$$((\varphi \circ \psi) \times (f_1 \circ \varphi \circ \psi) \times \cdots \times (f_{i-1} \circ \varphi \circ \psi) \times \pi_z^* \times (f_{i+1} \circ \varphi \circ \psi) \times \cdots \times (f_m \circ \varphi \circ \psi))$$

Where $\pi_z^*(x, z) := [1 : z]$. The unique point mapped to Q by Φ is $(0, 1/\lambda)$. Furthermore, a straightforward calculation shows that the following diagram commutes.

$$\begin{array}{ccc} \mathbb{A}^2 \supseteq W & \xrightarrow{\Phi} & \Phi(W) \subseteq D \\ \downarrow \psi & & \downarrow \pi \\ \mathbb{A}^2 & \xrightarrow{\varphi} & \mathbb{A}^2 \subseteq \mathbb{P}^2 \end{array}$$

We observe that $\Phi(W)$ is an open neighborhood of Q on D . It consists of the points that don't project to any of the other P_j (for $j \neq i$) and all but one of the points projecting to the point P_i (in the $(i+1)^{\text{st}}$ coordinate of L_i we do not reach $[0 : 1]$).

If D were a variety, we could obtain the desired curve as follows. Let X be the closure of $\pi^{-1}(C)$ in D . If we restrict f to the nonsingular points of C , f becomes an isomorphism with inverse π ; hence, the nonsingular points remain nonsingular, and C and X are birationally equivalent. Suppose Q is a point on X projecting to one of the singular points P_i of C . In that case, the above diagram shows that X has a neighborhood around Q , which is isomorphic to the affine blowup at P , and that in this isomorphism Q corresponds to one of the affine blowup points of P . Thus Q must be a smooth point on X . The map Φ reaches every point on D that could map to P_i except possibly one. Thus, if we choose coordinates of $\mathbb{P}^1$ such that this point does not lie on X , we see that $\Phi(W) \cap X$ is isomorphic to the open subset of the affine blowup of C at $(P_i)_*$, which excludes the finitely many points mapping to the other P_j ($j \neq i$). Thus exactly m_P points are mapped to P_i by π , and each such point corresponds to a tangent at P_i .

To complete the proof, we must show that D is a variety, i.e., irreducible. For this, we prove that D is the closure of the graph G , which we know to be a variety. Let S be any closed set in $\mathbb{P}^2 \times \mathbb{P}^1 \times \cdots \times \mathbb{P}^1$ containing G . We want to show that S also contains all the L_i . For simplicity, we show S contains the point Q from the above discussion, which was an arbitrary point on one of the L_i . From this it would follow that S contains D . The set $\Phi^{-1}(S)$ is closed in W , which contains $\Phi^{-1}(G) = W \setminus V(X)$. The latter set is dense, which implies $\Phi^{-1}(S) = W$ and, in particular, $Q \in S$. Lastly, the curve X is a closed curve in multi-projective space; however, we may consider it a projective curve via the Segre embedding. $\square$

4.3 Nonsingular Models

We now obtain a short proof of the main result we worked towards in Sections 3 and 4.

Proof of Theorem 4.0.1. By Corollary 2.3.2.1, we may project C to the plane to obtain a curve C' birationally equivalent to C . This curve C' may have nonordinary singularities; however, by Theorem 3.1.1, we may birationally transform this curve to another plane curve C'' , which only has ordinary singularities. By Theorem 4.2.1, we may birationally transform C'' to a curve X , which is smooth and projective. Thus, we obtain a birational map $f: X \rightarrow C$. It remains to show that f is a surjective morphism and that the pair (X, f) is unique. The fact that f is a morphism follows since X is smooth and C is projective (Lemma 2.3.3). For surjectivity, note that f is a nonconstant morphism between projective curves; thus, surjectivity is due to Lemma 2.3.3. Finally, let $g: Y \rightarrow C$ be another birational morphism from a smooth projective curve Y to C . Then $g^{-1} \circ f$ defines a birational map between smooth projective curves and thus an

isomorphism (Lemma 2.3.3). Thus, for any such pair (Y, g) , the morphism f factors over (Y, g) . $\square$

5 Divisors and $L(D)$ Spaces

In this section, we provide basic results about divisors and their $L(D)$ spaces. These two objects provide a systematic approach to studying functions with poles on algebraic curves, and therefore, they will play a central role in the Riemann-Roch theorem.

Throughout the rest of this thesis, let C be a projective algebraic curve, X its nonsingular model, K their function field, and $f: X \rightarrow C$ the (birational) morphism from X onto C . As we have seen in Section 3, up to a birational transformation, we may assume C to be planar with no nonordinary singularities. This assumption comes in handy in some of our proofs, although we will not make this assumption in general.

5.1 Divisors

As we have seen in Section 2, the only functions in K that are globally defined on X are constant functions; hence, to study more interesting functions, we must allow poles. A divisor will keep track of where these poles are and how bad they may be.

Definition 5.1.1. A divisor D on X is an element of the free abelian group over X ; in other words, it is a finite formal sum of points in X with integer coefficients. We shall call the free abelian group $\mathbb{Z}^{(X)}$ the divisor space over X .

The divisor space over X is naturally partially ordered by comparing coefficients at each point. We will use $\leq$ to denote this partial order and call a divisor D effective if $D \geq 0$. We would like to associate to any member $x \in K$ a unique divisor $\text{div}(x)$ that will encode the zeros and poles of x on X .

Lemma 5.1.2. *Let $x \in K$ be a nonzero function on X . The set of points S_1 in X , where $\text{ord}_P(x) > 0$ and the set of points S_2 in X where $\text{ord}_P(x) < 0$ are both finite sets.*

Proof. We show the claim only for S_2 . The claim follows for S_1 by considering $1/x$. First, we may cover X by affines and show that in each affine, S_1 is finite, so let U be an affine representative of X . The functions in K may all be represented as quotients f/g where $f, g \in \Gamma(U)$ and where $\Gamma(U)$ denotes the affine coordinate ring of U . It follows that for a point $P \in U$, the function x has a pole there if and only if in every representation $x = f/g$ we have $g(P) = 0$, and thus the pole set is precisely the algebraic set associated to the ideal $I_x \subseteq \Gamma(U)$ given by

$$I_x := \{G \in \Gamma(U) \mid x \cdot G \in \Gamma(U)\}.$$

This ideal is nonzero since we assume x to be nonzero. Thus, $V(I_x)$ defines a proper algebraic set of an algebraic curve and hence a finite collection of points. $\square$

Since X is a smooth algebraic curve, its points are in natural one-to-one correspondences with discrete valuation rings of K . Furthermore, by the previous lemma, any $x \in K^*$ has nonzero order in only finitely many places of X , so we may make the following definition:

Definition 5.1.3. Let $x \in K$ be a nonzero function on X . The divisor $\text{div}(x)$ is defined to be:

$$\text{div}(x) := \sum_{P \in X} \text{ord}_P(x)P.$$

We define the divisor of zeros of x , denoted by $(x)_0$, to be the divisor consisting of those points (along with their coefficients) where $\text{div}(x)$ has a positive coefficient. Similarly we may define x 's divisor of poles $(x)_\infty$ such that $\text{div}(x) = (x)_0 - (x)_\infty$.

We may observe that the map from $K^\times$ to $\mathbb{Z}^{(X)}$, mapping functions to their divisors, is a group homomorphism.

When C is planar, we may also assign divisors to other plane curves. These divisors encode the manner in which C intersects the other curves. In Section 2 we saw that for a form $G \in k[X, Y, Z]$, which does not have C as a component, it was possible to define $\text{ord}_P(G)$ at any $P \in X$ by considering the function G/L^m , where L is any line not through P and $\deg(G) = m$. This order is nonzero if and only if P is a place corresponding to an intersection point of C and G , which, in this case, is a finite set due to Bézout's theorem. Thus, it makes sense to use the following definition.

Definition 5.1.4. Suppose C is a projective plane curve with only ordinary singularities, and suppose $G \in k[X, Y, Z]$ is a form that defines a plane curve that does not have C as a component. Define the divisor $\text{div}(G)$ on X as:

$$\text{div}(G) := \sum_{P \in X} \text{ord}_P(G)P.$$

To any divisor D , we associate an integer $\deg(D)$, defined as the sum of those coefficients of points in D . Since D has finite support, this sum is always a finite sum. Furthermore, it is easily verified that $D \mapsto \deg(D)$ is a group homomorphism. As we shall see in the following lemma, the divisors that come from K^* are all of degree zero. Therefore, by the homomorphism theorem, they induce an equivalence relation on the space of divisors, which respects sums and taking degrees.

Lemma 5.1.5. Let $x \in K^\times$ be a nonzero function on X . Then, $\text{div}(x)$ is of degree zero.

Proof. We may write $x = g/h$, where g and h are residues of two homogeneous forms $G, H \in k[X, Y, Z]$ of equal degree, which both do not have C as a component (here we identified K with the function field of the planar representative C of X). Then, due to how we defined the order for forms, at any point $P \in X$, we have:

$$\text{ord}_P(x) = \text{ord}_P(G) - \text{ord}_P(H).$$

Therefore, globally we have

$$\text{div}(x) = \text{div}(G) - \text{div}(H).$$

By Bézout's theorem, $\text{div}(G)$ and $\text{div}(H)$ are divisors of equal degree. $\square$

Definition 5.1.6. A divisor is called principle if it may be represented as the divisor of a nonzero function on X . Two divisors D, D' are called linearly equivalent (written $D \equiv D'$) if they differ by a principle divisor.

According to the above comments, this notion of equivalence respects addition, taking degrees, and the zero class is that of principle divisors. Now, two plane curves have equivalent divisors if and only if they differ in a principle divisor, so when writing this principle divisor as a quotient of forms of equal degree, we get the following lemma.

Lemma 5.1.7. *Two plane curves G, G' have equivalent divisors if and only if there exist two forms A, B of equal degree such that:*

$$\text{div}(G) + \text{div}(A) = \text{div}(G') + \text{div}(B).$$

Now, take C again to be planar with only ordinary singularities. In general, C need not be smooth, and thus, the desingularization map $f: X \rightarrow C$ is not injective. The points on X , which are mapped to multiple points on C , are not intrinsic to X ; they depend on how we choose the plane curve C and the morphism f . For technical reasons, in later proofs, we would like to gain information about divisors on X by considering properties of C . It will be particularly useful to consider a projective plane curve C , which has only ordinary singularities. For any given X there could be many projective plane curves, with only ordinary singularities, which have X as nonsingular model. Thus we would like to keep track of the choice that we made in choosing C . Therefore we make the following definition:

Definition 5.1.8. Let C be a projective plane curve with only ordinary singularities, and let X be its nonsingular model. The adjoint divisor $E_{C,f}$ of C is the divisor on X defined as:

$$E_{C,f} := E := \sum_{Q \in X} (m_{f(Q)} - 1)Q.$$

Here $m_{f(Q)}$ denotes the multiplicity of the point $f(Q)$ on C . A plane curve G which does not contain C as a component is called adjoint to C if $\text{div}(G) \geq E$.

Note that $(m_{f(Q)} - 1) > 0$ only for those points that map to singular points on C . Since for every $P \in C$ exactly m_P points on X map to P , we find that the above sum is indeed a finite sum. Moreover, we may calculate the degree of E .

Lemma 5.1.9. *E is a divisor of degree $\sum_{P \in C} m_P(m_P - 1)$.*

Proof. A straightforward calculation yields:

$$\begin{aligned} \deg(E) &= \deg\left(\sum_{Q \in X} (m_{f(Q)} - 1)Q\right) \\ &= \sum_{Q \in X} (m_{f(Q)} - 1) \\ &= \sum_{P \in C} (m_P - 1) \cdot |\{f^{-1}(P)\}| \\ &= \sum_{P \in C} (m_P - 1) \cdot m_P. \end{aligned}$$

□

5.2 $L(D)$ Spaces

As indicated at the beginning of this section, our focus is on examining functions on X with specified poles. The following definition formalizes how we will do this.

Definition 5.2.1. Let D be a divisor on X , the space $L(D)$ is defined as:

$$L(D) = \{x \in K \mid x = 0 \text{ or } \operatorname{div}(x) + D \geq 0\}.$$

Since for any $f, g \in K^*$ and any $P \in X$, either $f + g = 0$ or $\operatorname{ord}_P(f + g) \geq \min\{\operatorname{ord}_P(f), \operatorname{ord}_P(g)\}$ holds, $L(D)$ is a k -vector space (note that we have $k \subseteq \mathcal{O}_P(X)^*$). We will denote by $l(D)$ the dimension of $L(D)$ as a k -vector space. The next few sections are dedicated to understanding how the space $L(D)$ behaves; this will culminate in Section 8, where we are finally able to calculate the dimension of $L(D)$ using the celebrated Riemann-Roch theorem.

In the next theorem, we begin our investigation by proving a few basic facts about these spaces.

Theorem 5.2.2. *Let D, D' be two divisors on X . Then the following statements hold.*

(1) *If $D \leq D'$, then $L(D) \subseteq L(D')$ and the inequality*

$$\dim_k(L(D')/L(D)) \leq \deg(D' - D)$$

holds.

(2) *$L(0) = k$ and $L(D) = 0$ if $\deg(D) < 0$.*

(3) $L(D)$ is finite-dimensional for all D and if $\deg(D) \geq 0$, then $l(D) \leq \deg(D) + 1$.

(4) If D and D' are equivalent then $l(D) = l(D')$.

Proof. (1) The first part is clear; since then, being in $L(D')$ is a weaker condition than being in $L(D)$. For the second statement, consider first $D' = D + P$. Choose some uniformizing parameter t at P and let m be the multiplicity in which P occurs in D' . Consider the map:

$$\Phi: L(D') \rightarrow k, \quad x \mapsto (t^m x)(P).$$

This map has kernel $L(D)$, thus we obtain

$$\dim_k(L(D')/L(D)) \leq 1.$$

For a general D' , we may repeatedly add single points to D in each step, increasing the dimension at most by one and the degree by exactly one.

(2) If $x \in L(0)$, then x has no poles. Thus, it is constant by the remarks in Section 2. If $\deg(D) < 0$, then $\operatorname{div}(x) + D$ can never be effective for $x \neq 0$, as taking the degree yields a negative integer.

(3) The claim is clear for $D = 0$, so assume this not to be the case. Since $\deg(D) \geq 0$, let P be some point in D with a positive coefficient and let $n := \deg(D)$. The divisor $D - (n+1)P$ has negative degree; therefore $l(D - (n+1)P) = 0$. Now, step by step, add P to this divisor, in each step increasing the degree by one and the dimension by at most one by part (1). We arrive at the original divisor after adding P to D exactly $n+1$ times.

(4) Suppose $D + \operatorname{div}(f) = D'$, where $f \neq 0$. Then consider the map:

$$\Phi_f: L(D') \rightarrow L(D), \quad x \mapsto f \cdot x.$$

Indeed Φ is k -linear and maps to $L(D)$, since for any $x \in L(D')$ we have $\operatorname{div}(\Phi(x)) = \operatorname{div}(x) + \operatorname{div}(f) \geq -D' + \operatorname{div}(f) = -D$. The map Φ also has an inverse map $\Phi_{1/f}$; thus, it is an isomorphism of k -vector spaces. $\square$

Corollary 5.2.2.1. *Let $D \leq D'$ be two divisors on X . Then*

$$l(D') \leq l(D) + \deg(D' - D).$$

Proof. By part (1) of Theorem 5.2.2 we know that

$$\dim_k(L(D')/L(D)) \leq \deg(D' - D).$$

Since $\dim_k(L(D')/L(D)) = l(D') - l(D)$ the result follows. $\square$

6 Riemann's Inequality and the Genus of a Curve

In the previous section, we introduced the concept of $L(D)$ spaces and derived an upper bound of $l(D)$ in Theorem 5.2.2. In this section, we will prove Riemann's inequality, giving a lower bound of $l(D)$.

A consequence of Riemann's inequality is that we will be able to define a nonnegative integer birational-invariant of X called the genus of X . In Section 10, we will see that the name genus is justified, as when $k = \mathbb{C}$, this corresponds to the topological genus that X carries as a smooth Riemann surface. In particular, the number we define as the genus may be considered a suitable generalization of the usual topological genus, which is only defined for smooth closed curves (as $\mathbb{R}$ or $\mathbb{C}$ manifolds).

The main result of this section is Riemann's inequality:

Theorem 6.0.1 (Riemann's Inequality). *Let X be the nonsingular model of an algebraic curve C . Then there exists an integer g such that for any divisor D on X we have:*

$$l(D) \geq \deg(D) + 1 - g.$$

The smallest such g is called the genus of X , the genus of C , or the genus of K .

The first part of this section is dedicated to proving Riemann's inequality. The main idea is to reduce Riemann's inequality to a statement about a very specific class of divisors using the residue theorem 6.1.1.

The second part of this section provides the tools to calculate the genus. As an interesting corollary, we will prove that algebraic curves of every genus exist.

6.1 Riemann's Inequality

For a certain class of divisors, $l(D)$ is easy to calculate, or we can at least prove the desired lower bound of Riemann's inequality more easily for them. The proof of Riemann's inequality relies on being able to pass from these special divisors to the general case. The next theorem will be the primary tool in doing so.

Theorem 6.1.1 (Residue Theorem). *Let C be a projective plane curve with only ordinary singularities, and let $D \equiv D'$ be two effective divisors on X . Let E be the adjoint divisor to C , and G be a form of degree m which is adjoint to C . Suppose we can express the divisor of G as*

$$\operatorname{div}(G) = D + E + A,$$

where $A \geq 0$. Then there exists an adjoint curve G' to C of degree m , such that

$$\operatorname{div}(G') = D' + E + A.$$

Proof. Let $C = V(F)$, where F is an irreducible form. Since $D \equiv D'$ we may write

$$\operatorname{div}(G) = D' + \operatorname{div}(f) + E + A.$$

Write $f = H/H'$ and then equivalently, we have

$$\operatorname{div}(GH') = D' + \operatorname{div}(H) + E + A \geq \operatorname{div}(H) + E.$$

Due to how divisors of plane curves were defined, we see that Noether's conditions in Definition 2.2.8 are satisfied for the forms GH' , H and F . Thus by Lemma 2.2.9 we may find forms G', F' such that

$$GH' = G'H + FF',$$

where $\deg(G') = m$. Then, calculating the divisor of G' , we find

$$\operatorname{div}(G') = \operatorname{div}(G) + \operatorname{div}(H'/H) = D' + E + A.$$

□

We will now give the first lower bounds of $l(D)$ for a special class of divisors. The proof follows the idea that can be found in Fulton [5]. However, we have modified it to avoid talking about the more general version of $L(D)$ spaces ($L^S(D)$ spaces) which are discussed there.

Theorem 6.1.2. *Let $x \in K \setminus k$ be a nonconstant function on X and $Z := (x)_0$ its divisor of zeros. Then the degree of Z is given by the degree $n := [K:k(x)]$ of the field extension $K/k(x)$. Furthermore there exists an integer τ such that for every integer r we have:*

$$l(rZ) \geq nr - \tau.$$

Proof. We prove the second claim first. Choose some integral basis $w_1, \dots, w_n$ of K over $k[x^{-1}]$ and let

$$w_i^{n_i} + a_{i,1}w_i^{n_i-1} + \dots + a_{i,n_i-1}w_i + a_{i,n_i} = 0 \quad (4)$$

be the integral equation of w_i where $a_{i,j} \in k[x^{-1}]$. Let S be the set where Z has a nonzero coefficient (equivalently positive). Since all $a_{i,j}$ are polynomials in x^{-1} , their only poles can lie at the zeros of x . Thus, $\operatorname{ord}_P(a_{i,j}) \geq 0$ whenever $P \notin S$. It follows that for each $P \notin S$ we must also have $\operatorname{ord}_P(w_i) \geq 0$ otherwise we would have $\operatorname{ord}_P(w_i^{n_i}) < \operatorname{ord}_P(w_i^{n_i-j}a_{i,j})$ for each $j > 0$. However, this can not occur since it would then be impossible to have Equation 4. Therefore, the only points where the w_i may have negative order lie in S . Hence, choosing t sufficiently large, we have $w_i + tZ \geq 0$ for all w_i . Then, for any r , the elements $w_i x^{-j}$ for $0 \leq j \leq r$, lie in the space $L((r+t)Z)$. They are also k -independent since the x^{-j} are k -linearly independent and the w_i are $k(x)$ -linearly independent. Thus we obtain the bound $l((r+t)Z) \geq n(r+1)$. Using Theorem 5.2.2, we obtain

$$n(r+1) \leq l((r+t)Z) = l(rZ) + \dim_k(L((r+t)Z)/L(rZ)) \leq l(rZ) + tm$$

where $m := \deg(Z)$, which shows

$$l(rZ) \geq nr + n - tm = nr - \tau,$$

where $\tau := tm - n$. To finish the proof, we still need to show $n = m$. Using Theorem 5.2.2, we have $l(rZ) \leq rm + 1$. Hence we obtain

$$nr - \tau \leq rm + 1,$$

for all $r \geq 0$, which is only possible if $n \leq m$.

We are left to prove the reverse inequality. Let V_1 denote the k -vector subspace of K consisting of functions with no poles at S . Similarly, define V_2 to be the k -vector subspace of K consisting of those functions $z \in K$ that satisfy

$$\text{ord}_P(z) \geq \text{ord}_P(x) \text{ for all } P \in S.$$

In general, both spaces are infinite-dimensional since we have no restrictions on what happens outside of the set S . However, we do know that $V_2 \subseteq V_1$. Therefore, it makes sense to consider the quotient space V_1/V_2 . We claim that it is finite-dimensional and of dimension m .

We prove this by induction over m . Suppose $Z = P_1 + \cdots + P_m$ (where points may occur multiple times in the above sum). If $m = 1$, then $Z = P_1$ and the evaluation map

$$\Phi_P: V_1 \rightarrow k, f \mapsto (f)(P),$$

has kernel V_2 . The evaluation map is also surjective as we may choose a function that has P as a simple pole (take, for example, $1/t$). So for $m = 1$, we obtain the result. If $m > 1$, consider the divisor $Z' := P_1 + \cdots + P_{m-1}$ and let V'_2 be its corresponding vector space (defined in the same way that V_2 was). By induction, we know that:

$$\dim_k(V_1/V'_2) = m - 1.$$

Thus it suffices to prove

$$\dim_k(V'_2/V_2) = 1.$$

Indeed, suppose α is the coefficient of P_m in Z' and that t is a uniformizing parameter of $\mathcal{O}_P(X)$. The evaluation map $\Phi_{\alpha,P}$ defined as

$$\Phi_{\alpha,P}: V'_2 \rightarrow k, f \mapsto (ft^{-\alpha})(P),$$

has kernel V_2 . We claim it is also surjective. Indeed, take a projective plane curve C , birational to X , on which the points in S correspond to smooth points. Then, for each P_i , we may choose a line L_i , which intersects C transversely at P_i and passes through none of the other P_j . We may also choose a line L , which passes through none of the P_i . Then we have

$$\Phi_{\alpha,P} \left(\frac{\prod_{i=1}^{m-1} L_i}{L^{m-1}} \right) \neq 0,$$

which shows surjectivity and, thus, that the dimension of V_1/V_2 is m .

Choose m vectors $v_1, \dots, v_m$ in V_1 , which map to a basis in V_1/V_2 . To show that $m \leq n = [K:k(x)]$, it suffices to prove that the v_i are $k(x)$ -linearly independent as well. Indeed, if this were not the case, we would have a nontrivial linear combination

$$\lambda_1 v_1 + \dots + \lambda_m v_m = 0, \text{ with } \lambda_i \in k(x)$$

By clearing out a common denominator, all the λ_i may be chosen to be polynomials in x . Since not all of the polynomials are zero, we can divide by the greatest common factor to assume that not all λ_i are divisible by x . This leads to a contradiction since it would yield the equation

$$\begin{aligned} 0 &= \lambda_1 v_1 + \dots + \lambda_m v_m \\ &= (x f_1(x) + \mu_1) v_1 + \dots + (x f_m(x) + \mu_m) v_m, \end{aligned}$$

where $f_i \in k[x]$, $\mu_i \in k$, and not all $\mu_i = 0$. Rearranging the terms divisible by x to the left leads to a nontrivial linear combination of the v_i in V_1/V_2 . $\square$

As a direct consequence of Theorem 6.1.2, we get Riemann's inequality.

Proof of Riemann's inequality 6.0.1. The theorem is equivalent to the fact that quantity

$$s(D) := \deg(D) + 1 - l(D)$$

is universally bounded from above for all divisors D on X . First, note that the above quantity is the same for any two equivalent divisors since each of the terms is. Furthermore, by Corollary 5.2.2.1, $S(D)$ is increasing in D . That is, if $D \leq D'$ holds, then $S(D) \leq S(D')$ also must hold. Let $x \in K \setminus k$ be a nonconstant function and let $Z := (x)_0$ be its divisor of zeros. By Theorem 6.1.2, we know that there exists an integer g^* such that $s(nZ) \leq g^*$ for all $n \in \mathbb{Z}$, in particular $g^* \geq S(0 \cdot Z) = 0$, so g^* is nonnegative. Let g be the smallest such integer, and let D be any divisor on X . We will also show that $s(D) \leq g$ holds.

By the above comments, it suffices to pass to a divisor D' which is equivalent to D and for which there exists some integer n such that $D' \leq nZ$, for then we have $s(D) = s(D') \leq s(nZ) \leq g$ as desired. To prove that there exists such a divisor, first write $D = n_1 P_1 + \dots + n_t P_t$ and $Z = m_1 Q_1 + \dots + m_s Q_s$. Since Z is effective, Z has a coefficient at least as large as that of D at any point where D has a nonpositive coefficient. Whenever D and Z both have positive coefficients, scaling Z sufficiently large ensures that Z has a larger coefficient there. Thus, we need only change D at those points where D has a positive coefficient and Z has a zero coefficient; we call such points bad. Let P be a bad point of D then we have $\text{ord}_P(x) \leq 0$, thus $\text{ord}_P(1/x) \geq 0$. Therefore, $1/x$ takes on a well-defined value $\alpha \in k$ at P . Then the function $1/x - \alpha$ has positive order here and $\text{ord}_P((1/x - \alpha)^{m_P}) \geq m_P$. Furthermore, this function only has poles at those points where Z has a positive coefficient, so the divisor $D - \text{div}((1/x - \alpha)^{m_P})$ has one less bad point. Repeating this process for all bad points, we get a divisor equivalent to D with no bad points. $\square$

The Riemann-Roch theorem tells us, for any divisor D , the precise dimension of $L(D)$. Using Riemann's inequality, we can now prove the Riemann-Roch theorem for all sufficiently large divisors. This will be a crucial step in proving the full theorem in Section 8.

Corollary 6.1.2.1 (Riemann-Roch Theorem for Large Divisors). *There exists an integer N such that for any divisor D of degree $\deg(D) \geq N$, we have*

$$l(D) = \deg(D) + 1 - g.$$

Proof. Since g was defined to be the least integer such that $l(D) \geq \deg(D) + 1 - g$ holds for all D , there must exist some divisor D_0 such that $l(D_0) = \deg(D_0) + 1 - g$. Define $N := \deg(D_0) + g$. Suppose D is any divisor of degree at least N then by Riemann's inequality 6.0.1 we know that:

$$l(D - D_0) \geq \deg(D - D_0) + 1 - g > 0,$$

thus there exists some nonzero function $f \in L(D - D_0)$. By definition, we then have $\text{div}(f) + D \geq D_0$, so D is equivalent to a divisor that is larger than D_0 . Using the notation as in the proof of Riemann's inequality, we then have $g \geq s(D) \geq s(D_0)$. By choice of the divisor D_0 we have $s(D_0) = g$, which implies $s(D) = g$. $\square$

6.2 The Genus Formula

In Riemann's inequality 6.0.1, we have proven the existence of the genus and seen that it was a birational invariant of X since it depended only on properties of the function field. We also find that the genus is a nonnegative integer from the proof of Theorem 6.0.1. However, we have not yet seen how to calculate it. In the following theorem, we provide a method of calculating the genus.

Theorem 6.2.1 (The Genus Formula). *Let C be a projective plane curve with only ordinary singularities, and let F be an irreducible form of degree n , which defines C . For any $P \in C$, let $m_P \geq 0$ be the multiplicity in which it occurs on C . Then, the genus of C (or of X) is given by the formula*

$$g = g^* := \frac{(n-1)(n-2)}{2} - \sum_{P \in C} \frac{m_P(m_P-1)}{2}.$$

Proof. We find a family of divisors $(D_i)_{i \in I}$ containing divisors of arbitrary large degree for which:

$$l(D_i) = \deg(D_i) + 1 - g^* \text{ (for all } i \in I\text{)}.$$

The claim then follows by Corollary 6.1.2.1. First, by Corollary 3.2.3.2 choose coordinates such that the line $Z = 0$ intersects C transversely in n distinct points $P_1, \dots, P_n$. Let E be the adjoint divisor to C and let E_m be the divisor defined by the equation:

$$E_m := m \sum_{i=1}^n P_i - E.$$

Since E is of degree $\sum_{P \in C} m_P(m_P - 1)$, E_m has degree $mn - \sum_{P \in C} m_P(m_P - 1)$. We seek to calculate the dimension of $L(E_m)$, and for this, we write $L(E_m)$ as the cokernel of two spaces of which we may easily calculate the dimension.

Let V_m be the projective space of forms of degree m which intersect C at every point $P \in C$ in multiplicity at least $m_P - 1$. From Lemma 3.3.1, if m is sufficiently large, the projective dimension of V_m is given by

$$\frac{m(m+3)}{2} - \sum_{P \in C} \frac{m_P(m_P - 1)}{2}.$$

Therefore, adding one to obtain the dimension as a k -vector space, we conclude that V_m has dimension

$$\frac{m(m+3)+2}{2} - \sum_{P \in C} \frac{m_P(m_P - 1)}{2} = \frac{(m+1)(m+2)}{2} - \sum_{P \in C} \frac{m_P(m_P - 1)}{2}.$$

Furthermore, let W_{m-n} (again we assume m to be sufficiently large) be the vector space of forms of degree $m - n$. A k -basis of W_{m-n} is given by the monomials:

$$\{X^i Y^j Y^k \mid i, j, k \in \mathbb{N}, i + j + k = m - n\}$$

of which there are

$$\begin{aligned} \sum_{i=0}^{m-n} \left(\sum_{j=0}^{m-n-i} 1 \right) &= \sum_{i=0}^{m-n} (m - n - i + 1) \\ &= (m - n)(m - n + 1) - \frac{(m - n)(m - n + 1)}{2} + (m - n + 1) \\ &= \frac{(m - n + 1)(m - n + 2)}{2}. \end{aligned}$$

Finally, consider the sequence.

$$0 \longrightarrow W_{m-n} \xrightarrow{f} V_m \xrightarrow{h} L(E_m) \longrightarrow 0,$$

where we define the maps as

$$\begin{aligned} f: W_{m-n} &\rightarrow V_m, \quad H \mapsto F \cdot H \\ h: V_m &\rightarrow L(E_m), \quad H \mapsto \frac{\overline{H}}{(\overline{Z})^m}. \end{aligned}$$

We claim that this is an exact sequence. First, note that f maps to the desired space, as at any point $m_P(FH) = m_P(F) + m_P(H)$. Similarly, h maps to the desired space. Furthermore, it is apparent that f is injective. Now, $\overline{H}/\overline{Z}^m = 0$ occurs if and only if H is divisible by F , so we get exactness at V_m . It remains to show that h is surjective; indeed, let $x = \frac{\overline{A}}{\overline{B}} \in L(E_m)$. Then, by definition,

$$\operatorname{div}(A) - \operatorname{div}(B) \geq E - m \sum_{i=1}^n P_i$$

or equivalently

$$\operatorname{div}(AZ^m) \geq E + \operatorname{div}(B).$$

Therefore, Noether's conditions are satisfied, and by Lemma 2.2.9, we have an equation:

$$AZ^m = GF + BH,$$

or equivalently

$$\frac{H}{Z^m} = \frac{A}{B} - F \frac{G}{BZ^m}.$$

Thus, $h(H) = x$ as desired. Hence, we may calculate the dimension of $L(E_m)$ as:

$$\begin{aligned} l(E_m) &= \dim_k(V_m) - \dim_k(W_{m-n}) \\ &= \left(\frac{(m+1)(m+2)}{2} - \sum_{p \in C} \frac{m_P(m_P-1)}{2} \right) - \frac{(m-n+1)(m-n+2)}{2} \\ &= \left(mn - \sum_{p \in C} m_P(m_P-1) \right) + 1 - \left(\frac{(n-1)(n-2)}{2} - \sum_{P \in C} \frac{m_P(m_P-1)}{2} \right) \\ &= \deg(E_m) + 1 - g^*. \end{aligned}$$

Since $\deg(E_m)$ tends to infinity as m does, the result follows from Corollary 6.1.2.1. $\square$

Remark. When C is smooth of degree n , this becomes the usual well-known genus formula:

$$g = \binom{n-1}{2}.$$

Corollary 6.2.1.1. *Let C be a plane curve of degree n , then*

$$g \leq g^* := \frac{(n-1)(n-2)}{2} - \sum_{P \in C} \frac{m_P(m_P-1)}{2}.$$

Proof. If C has only ordinary singularities, then the previous theorem shows these quantities are equal. If C has nonordinary singularities, then using Theorem 3.1.1, we can transform C birationally to a planar curve with only ordinary singularities in such a way that decreases g^* (not necessarily strictly decreasing). $\square$

Using the notation as in the above proof, we may use the fact that h was surjective to get the following corollary.

Corollary 6.2.1.2. *Suppose $f \in L(E_m)$. Then f may be written as the residue of the quotient of $\frac{G}{Z^m}$ where G is an adjoint curve to C of degree m .*

Another important result that we obtain from 6.2.1 is that there are curves of every genus.

Corollary 6.2.1.3. *Let $d \geq 0$ be any integer. Then, there exists a planar curve of genus d .*

Proof. We prove the result only for $\text{char}(k) \neq 2$. Let $\beta_1, \dots, \beta_d, \alpha_1, \dots, \alpha_{d+2}$ be a collection of pairwise distinct elements in k and consider the homogeneous form $F \in k[X, Y, Z]$ defined by

$$F(X, Y, Z) = Y^2 \cdot \prod_{i=1}^d (X - Z\beta_i) + \prod_{i=1}^{d+2} (X - Z\alpha_i).$$

First we claim that this polynomial is irreducible; indeed, consider it a polynomial over the field $k(X, Z)$ in the indeterminate Y . The polynomial

$$Y^2 + \frac{\prod_{i=1}^{d+2} (X - Z\alpha_i)}{\prod_{i=1}^d (X - Z\beta_i)}$$

is associated to F in $k(X, Z)[Y]$ and irreducible since $-\frac{\prod_{i=1}^{d+2} (X - Z\alpha_i)}{\prod_{i=1}^d (X - Z\beta_i)}$ is not a square in $k(X, Z)$. The field $K(X, Z)$ is the field of fractions of $k[X, Z]$. Therefore, since $F(X, Y, Z)$ is primitive when viewed as a polynomial over $k[X, Z]$, Gauss's lemma implies that F is irreducible in $F[X, Y, Z]$ as well.

Next we compute the singularities of F on the affine $Z \neq 0$, where we have:

$$F_Y = 2Y \cdot \prod_{i=1}^d (X - Z\beta_i)$$

from which it follows that any affine singularities lie on $V(Y)$. Indeed, if for such a singular point $P = [x:y:1]$ we have $y \neq 0$, the above equation implies $x = \beta_i$ for some i , which contradicts $P \in V(F)$. Now, the line $V(Y)$ intersects $V(F)$ in the $d+2 = \deg(F)$ distinct points $[\alpha_i:0:1]$, thus by Bézout's theorem none of these points can be singular. In particular, the singularities lie on the affine $Y \neq 0$ and $V(Z)$. At such points, we find

$$F_Y(P) = 2x^d.$$

This means we can only have a singular point at $[0:1:0]$. Dehomogenizing F at Y , the defining form becomes

$$F(X, Y, Z)_* = \prod_{i=1}^d (X - Z\beta_i) + \prod_{i=1}^{d+2} (X - Z\alpha_i)$$

which has multiplicity d at $(0,0)$, with the d pairwise distinct tangents given by $V(X - Z\beta_i)$. Finally, we may use Theorem 6.2.1 to calculate the genus g of $V(F)$ as

$$g = \frac{(d+1)d}{2} - \frac{d(d-1)}{2} = d.$$

□

7 Canonical Divisors

In this section, we introduce differentials and canonical divisors. When our base field k corresponds to the complex numbers, the nonsingular model X becomes a Riemann surface, which naturally carries with it its space of differentials. In the holomorphic case, this space has proven useful in calculating the dimension of $L(D)$, thus it is natural to extend these notions to the more general fields we work with here.

7.1 Kähler Differentials.

In the following, we would like to assign differentials to elements of a domain $R \supseteq k$, where the ring R will usually be interpreted as some ring of functions taking values in $k \cup \{\infty\}$. To do this, we simply take the properties of differentials in the analytic case as a definition.

Definition 7.1.1. Let R be a domain containing k and F be the free R -module over R . For any $x \in R$, denote by $[x]$ its corresponding element in F . Finally, let N be the submodule of F generated by the following relators

- $[x] + [y] - [x + y]$
- $[\lambda x] - \lambda[x]$ for $\lambda \in k$
- $[x \cdot y] - (x[y] + y[x])$ (Leibniz rule).

The R -module $\Omega_k(R) := F/N$ is called the space of differentials of R over k ; its elements are called differentials; and for any $x \in R$, we denote by dx its corresponding differential in $\Omega_k(R)$.

Remark. $\Omega_k(R)$ is a k -vector space by restriction of scalars.

The R -module $\Omega_k(R)$ satisfies a universal property, which is often more useful to work with than the explicit definition given above. To state this property, recall the definition of a derivation.

Definition 7.1.2. Let M be a R -module. A derivation over k is a map $\Phi: R \rightarrow M$ satisfying:

- $\Phi: R \rightarrow M$ is a k -linear map

- $\Phi(xy) = xd(y) + yd(x)$.

One may verify that the map $d: R \rightarrow \Omega_k(R), x \mapsto dx$ is a derivation. A routine check shows that even more is true:

Lemma 7.1.3 (Universal Property of Kähler Differentials). *Let $R \supseteq k$ be a domain with corresponding space of differentials $\Omega_k(R)$. Then, for any R -module M and any derivation $\Phi: R \rightarrow M$ there exists a unique R -module homomorphism $\Phi': \Omega_k(R) \rightarrow M$ such that the following diagram commutes:*

$$\begin{array}{ccc} R & \xrightarrow{\Phi} & M \\ \downarrow d & \nearrow \exists! \Phi' & \\ \Omega_k(R) & & \end{array}$$

That is, we have $\Phi = \Phi' \circ d$. Furthermore, any other R -module with this property is uniquely isomorphic to $\Omega_k(R)$.

Since the elements $\{dx : x \in R\}$ generate $\Omega_k(R)$ as an R -module, we can express every differential as an R -linear combination of these elements. Furthermore, due to the Leibniz rule and k -linearity, we also see that if we choose $x_1, \dots, x_n$ from R , the differential of any polynomial in the x_i may be calculated as an R -linear combination of $dx_1, \dots, dx_n$. In particular, if R is a finitely generated k -algebra with generators $x_1, \dots, x_n$, then $\Omega_k(R)$ is generated as an R -module by $dx_1, \dots, dx_n$. Thus, if R is a field, we obtain an upper bound on the dimension of $\Omega_k(R)$ as a R -vector space in this way.

In general, R need not be a field; however, since R is an integral domain, we can form its field of fractions K . The spaces $\Omega_k(R)$ and $\Omega_k(K)$ compare in a natural way, as seen in the following lemma.

Lemma 7.1.4. *Let $R \supseteq k$ be a domain and K its field of fractions. Then*

$$\Omega_k(K) \simeq \Omega_k(R) \otimes_R K.$$

Proof. We verify that $\Omega_k(K)$ satisfies the universal property of the tensor product. Define the map

$$\begin{aligned} T: \Omega_k(R) \times K &\rightarrow \Omega_k(K) \\ (\omega, x) &\mapsto x \cdot \omega, \end{aligned}$$

where we may identify ω with an element of $\Omega_k(K)$ by Lemma 7.1.3, choosing $M = \Omega_k(K)$. Suppose A is any R -module and we have some R -bilinear map

$$b: \Omega_k(R) \times K \rightarrow A.$$

By the comments above $\Omega_k(K)$ is generated as a K -vector space by elements $d_z, z \in K$; however, since every $z \in K$ may be represented as a quotient x/y of elements in R and the Leibniz rule applies:

$$dz = d(x/y) = y^{-1}d(x) + xd(1/y) = y^{-1}d(x) - xy^{-2}d(y).$$

We find that $\Omega_k(K)$ is already K -generated by elements $dx, x \in R$. Now, for such elements, the unique way to define a homomorphism $b': \Omega_k(K) \rightarrow A$, such that the below diagram commutes, is $b'(\lambda dx) = b(\lambda, dx)$.

$$\begin{array}{ccc} \Omega_k(R) \times K & \xrightarrow{b} & A \\ \downarrow T & \nearrow b' & \\ \Omega_k(K) & & \end{array}$$

Extending this map linearly to all of $\Omega_k(K)$ yields the desired result. $\square$

Remark. In the above proof, we omitted tedious yet routine arguments, which guaranteed that the maps that we defined were well-defined. This comes down to repeatedly leveraging Lemma 7.1.3.

Corollary 7.1.4.1. *Let $R \supseteq k$ be a domain and K its field of fractions, any derivation $\Phi: R \rightarrow M$ where M is a K -vector space extends uniquely to a derivation on K .*

Proof. Since M is a K -vector space $M \simeq M \otimes_K K$. By Lemma 7.1.3, there is a unique R -module homomorphism $\Phi': \Omega_k(R) \rightarrow M$ corresponding to Φ . The unique extension is given by the map corresponding to

$$\Phi' \otimes_R \text{Id}: \Omega_k(R) \otimes_R K \rightarrow M \otimes_R K.$$

$\square$

We are particularly interested in the case where K corresponds to the function field of an algebraic curve. In this special case, we are able to calculate the dimension of $\Omega_k(K)$ as a K -vector space.

Theorem 7.1.5. *Let K be a one-dimensional algebraic function field over k . Then $\Omega_k(K)$ is a one-dimensional K -vector space. Furthermore, if $x \in K \setminus k$ is such that K is a separable extension over $k(x)$, then dx is a basis for $\Omega_k(K)$.*

Before we prove the statement, note that the second condition is always guaranteed when $\text{char}(k) = 0$. Furthermore, as we have seen in Lemma 2.2.1, when $\text{char}(k) = p$, it is still possible to choose x in such a way that the condition holds. More generally, any nonzero $\omega \in \Omega_k(K)$ will be a basis for this space, so the second part comes down to checking $dx \neq 0$.

Proof. Let $F \in k[X, Y]$ be an irreducible polynomial defining an affine curve with function field K (Lemma 2.3.2.1) so $K = k(x, y)$ where $k[x, y] := k[X, Y]/(F)$. F_X and F_Y cannot both be zero. This is clear when $\text{char}(k) = 0$. In $\text{char}(k) = p > 0$, having vanishing derivatives means that both variables occur only in powers of p . Then, using the Frobenius endomorphism, we may write

$$F(X, Y) = F'(X^P, Y^P) = G(X, Y)^P,$$

where G is the polynomial whose coefficients correspond to p 'th roots of those of F' . This contradicts the irreducibility of F . Thus, without loss of generality, suppose $F_Y \neq 0$. This is both equivalent to $F_Y \notin (F)$ and $F_Y(x, y) \neq 0$. Using the fact that $F(x, y) = 0$, its differential is as well, so we compute

$$0 = d(F) = F_X(x, y)dx + F_Y(x, y)dy,$$

which yields

$$dy = -\frac{F_X(x, y)}{F_Y(x, y)}dx,$$

which shows that dy lies in the K -span of dx . Since the differential of any rational function in x, y may be expressed as a K -linear combination of dx and dy , we have shown $\dim_K(\Omega_k(K)) \leq 1$. Define $u := -\frac{F_X(x, y)}{F_Y(x, y)}$ to be the term above and consider the map

$$\Phi : K[X, Y] \rightarrow K, \quad G \mapsto G_X - uG_Y.$$

This map is clearly k -linear, and it is also a derivation since

$$\begin{aligned} \Phi(GH) &= (GH)_X - u(GH)_Y \\ &= G_X \cdot H + H_X \cdot G - uG_Y \cdot H - uH_Y \cdot G \\ &= (G_X - u \cdot G_Y)H + (H_X - u \cdot H_Y)G \\ &= \Phi(G)H + \Phi(H)G. \end{aligned}$$

Lastly, since $\Phi(F) = F_X - uF_Y = 0$, this derivation induces a well-defined derivation on $k[x, y]$, which in turn extends to a unique k -linear map from $\Omega_k(K)$ to K . Since $\Phi(X) = 1$, we must have that $\Omega_k(K) \neq 0$, which shows the first part. For the second part, let $x \in K$ such that K is a separable extension over $k(x)$. If $K = k(x)$, we are done, since if $dx = 0$, then so would the differential of any rational function in x , which contradicts $\Omega_k(K) \neq 0$. Otherwise, we may choose a primitive element $y \in K \setminus k(x)$ such that $K = k(x, y)$ since $K/k(x)$ is a separable extension. Let $F(x, Y)$ be the minimal polynomial of y over $k(x)$, whereby clearing out denominators, we may assume F has coefficients only in $k[x]$. Since by assumption K is separable over $k(x)$, we must have $F_Y \neq 0$. Then, we may proceed as before. $\square$

With Theorem 7.1.5, we can define derivatives with respect to parameters.

Definition 7.1.6. Let K be an algebraic function field in one variable and let $t \in K$ such that dt is a basis of $\Omega_k(K)$. Then, for any $x \in K$, there exists a unique function $\lambda \in K$ such that $dx = \lambda dt$. This scalar is called the derivative of x with respect to t and is denoted $\frac{dx}{dt}$.

7.2 Canonical Divisors

Definition 7.1.6 gains importance when we want to define the order of a differential at a point on X because it allows us to translate this to the already defined concept of the order of a function. We will need to use the following facts to make a precise definition.

Lemma 7.2.1. *Let K be an algebraic function field in one variable and suppose $\mathcal{O} \supseteq k$ is a discrete valuation ring of K with uniformizing parameter t . Then, dt is a basis for $\Omega_k(K)$.*

Remark. By Theorem 7.1.5, the above lemma is immediate in $\text{char}(k) = 0$.

Proof. It suffices to show that $dt \neq 0$. Since $\mathcal{O}$ is a discrete valuation ring, every element $f \in \mathcal{O}$ corresponds to a unique power series over k where the first n coefficients are determined by the unique $\lambda_0, \dots, \lambda_{n-1} \in k$ such that

$$f = \lambda_0 t^0 + \dots + \lambda_n t^n + t^n g \quad (\text{for some } g \in \mathcal{O}).$$

The map h taking f to its power series is a ring homomorphism, and the map taking a power series to its formal derivative is a derivation. So by Lemma 7.1.3 we get a unique map $\Phi: \Omega_k(\mathcal{O}) \rightarrow k[[T]]$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O} & \xrightarrow{h} & k[[T]] \xrightarrow{D} k[[T]] \\ \downarrow d & & \nearrow \Phi \\ \Omega_k(\mathcal{O}) & & \end{array}$$

Now, $(D \circ h)(t) = 1$ and therefore, $\Phi(dt) = 1$, which implies $dt \neq 0$. Since K is $\mathcal{O}$'s field of fractions, the result follows from Corollary 7.1.4.1. $\square$

In the analytic case, if a meromorphic function $f \in K$ does not have a pole at a particular point on the manifold X , then no matter how we choose charts of X , the derivative does not either. The following generalizes this.

Theorem 7.2.2. *Let K be a one-dimensional algebraic function field over k , $\mathcal{O} \supseteq k$ a discrete valuation ring of K , and t a uniformizing parameter of $\mathcal{O}$. If $f \in \mathcal{O}$, then $\frac{df}{dt} \in \mathcal{O}$ as well.*

Proof. We may choose an affine curve represented by an irreducible polynomial $F \in k[X, Y]$, which has function field K and for which $\mathcal{O} = \mathcal{O}_P(F)$, where $P = (0, 0)$. Then, K is the quotient field of the integral domain

$$k[x, y] := k[X, Y]/(F),$$

thus, we may write it as $K = k(x, y)$. We choose a nonnegative integer N large enough such that $\text{ord}_P(\frac{dx}{dt}) \geq -N$ and $\text{ord}_P(\frac{dy}{dt}) \geq -N$. Then, for any polynomial $g \in k[x, y] \subseteq \mathcal{O}_P$ we also have $\text{ord}_P(\frac{dg}{dt}) \geq -N$ since $dg = g_x \cdot dx + g_y \cdot dy$ thus, $\frac{dg}{dt} = g_x \cdot \frac{dx}{dt} + g_y \cdot \frac{dy}{dt}$ which has order at least $-N$. Since any $f \in \mathcal{O}_P$ may

be represented as a quotient of two polynomials $f = h/g$ where $g(P) \neq 0$ we calculate

$$\begin{aligned}
df &= d(h/g) = g^{-1} \cdot dh + h \cdot d(g^{-1}) \\
&= g^{-1} \cdot dh - hg^{-2} \cdot dg \\
&= g^{-2}(g \cdot dh - h \cdot dg). \\
&= g^{-2} \left(g \cdot \frac{dh}{dt} dt - h \cdot \frac{dg}{dt} dt \right) \\
&= g^{-2} \left(g \cdot \frac{dh}{dt} - h \cdot \frac{dg}{dt} \right) dt
\end{aligned}$$

Thus, computing the order at P

$$\begin{aligned}
\text{ord}_P \left(\frac{df}{dt} \right) &= \text{ord}_P \left(g^{-2} \left(g \cdot \frac{dh}{dt} - h \cdot \frac{dg}{dt} \right) \right) \\
&= \text{ord}_P \left(g \cdot \frac{dh}{dt} - h \cdot \frac{dg}{dt} \right) \\
&\geq -N.
\end{aligned}$$

This shows, that the derivative of any element in $\mathcal{O}$ with respect to t , has order at least $-N$. But now, since $h \in \mathcal{O}_P$, we may consider f 's series expansion in t up to order $N-1$ and write

$$f = \lambda_0 t^0 + \lambda_1 t^1 + \cdots + \lambda_{N-1} t^{N-1} + t^N G,$$

where the $\lambda_i \in k$ and $G \in \mathcal{O}_P$. Since $d(t^n) = nt^{n-1}dt$ we see that $\frac{df}{dt} \in \mathcal{O}_P$. $\square$

Using Lemma 7.2.1, we can define the order of differentials at points, which is the final component we need to define canonical divisors.

Definition 7.2.3. Let $\omega \in \Omega_k(K)$ be a nonzero differential on X and $P \in X$ be a point on X . Furthermore, let $t \in \mathcal{O}_P$ be a uniformizing parameter at P . Then, the order of ω at P is defined as the order of the unique function $\lambda \in K$ such that $\omega = \lambda \cdot dt$. We denote it by

$$\text{ord}_P(\omega) := \text{ord}_P \left(\frac{\omega}{dt} \right) := \text{ord}_P(\lambda).$$

Remark. A priori, the above definition depends on the choice of uniformizing parameter t . However, if t' is another uniformizing parameter, then by Theorem 7.2.2, both $\frac{dt}{dt'}$ and $\frac{dt'}{dt} = 1/(\frac{dt}{dt'})$ have nonnegative order at P . Hence, they have order 0 at P . Since $\frac{\omega}{dt} = \frac{\omega}{dt'} \cdot \frac{dt'}{dt}$, this order is indeed independent of choice of parameter.

Definition 7.2.4. Let $\omega \in \Omega_k(K)$ be a nonzero differential on X . The divisor of ω is defined to be

$$\text{div}(\omega) := \sum_{P \in X} \text{ord}_P(\omega).$$

A divisor, which may be written as the divisor of a differential, is called a canonical divisor.

The above definition does not guarantee that the divisor of a differential is, in fact, a divisor in the sense of Definition 5.1.1, since for it to lie in divisor space, we need almost all the coefficients in the above sum to vanish. This is, however, the case, as we shall see in Corollary 7.2.6.1. Taking this fact for granted for a moment, we may collect a few facts about canonical divisors in the following lemma.

Remark. An alternative approach can be taken to handle concerns about circular logic when assuming that canonical divisors have finite support. Instead of focusing solely on the space of divisors, we can consider canonical divisors as elements of the free R -Product over X , which is a larger space. Canonical divisors are part of this bigger R -module, and the regular divisors can also be included in this module. Furthermore, the usual concept of equivalence can also be applied to this module. Ultimately, Corollary 7.2.6.1 demonstrates that canonical divisors are indeed contained within the smaller submodule consisting of the regular divisors.

Lemma 7.2.5. *Any two canonical divisors are equivalent, and a divisor equivalent to a canonical divisor is also canonical.*

Proof. Let ω_1, ω_2 be nonzero differentials in $\Omega_k(K)$. Since $\Omega_k(K)$ is a one-dimensional K -vector space there exists a function $f \in K$ such that $\omega_1 = f\omega_2$. Then $\text{div}(\omega_1) = \text{div}(f) + \text{div}(\omega_2)$. Conversely, if D is a divisor, which may be expressed as $D = \text{div}(f) + \omega$, then $D = \text{div}(f\omega)$ is also canonical. $\square$

Remark. The above lemma shows that the set of canonical divisors forms an equivalence class in the space of divisors, and thus, any two canonical divisors have the same degree.

In the following theorem, we explicitly calculate a canonical divisor. In particular, we show that Definition 7.2.4 is a justified definition. In general, we follow the idea of the proof, which can be found in Fulton [5]. However, we have modified it at certain parts such that the arguments hold cleanly in $\text{char}(k) = p > 0$ as well. In particular, we pay special attention to the assumptions about how C may be placed in the plane.

Theorem 7.2.6. *Let C be a projective plane curve with only ordinary singularities of degree $n \geq 3$, and let X be its nonsingular model. Let E be the adjoint divisor to C and let G be any form of degree $n - 3$. Then $\text{div}(G) - E$ is a canonical divisor.*

Proof. By Corollary 3.2.3.2, we may change to coordinates X, Y, Z such that $V(Z)$ intersects C transversely in points $P_1, \dots, P_n$ (in particular they are smooth on C). Since C has only finitely many multiple points, each with finitely many tangents, we may find two points on Z which lie on none of these tangents and also do not lie on C , and choose them as $[1 : 0 : 0]$ and $[0 : 1 : 0]$. Proving the statement for this transformed curve is equivalent to the base curve, as these transformations leave E unchanged and map G to another form of degree $n - 3$ again.

Let $F(X, Y, Z)$ be the (irreducible) form defining C with partial derivatives F_X, F_Y, F_Z . If $\text{char}(k) = 0$, then none of the partial derivatives are zero, or else F would be a form in fewer than three variables and would thus factor into a product of lines. Since F is irreducible, this would imply that F itself defines a line, which contradicts the degree of F being at least three. Consider the case when $\text{char}(k) = p$, where we claim $F_X \neq 0$. Indeed, consider the tangent of F at P_1 . Since P_1 is simple on F , L is of the form

$$L = F_X(P_1)X + F_Y(P_1)Y + F_Z(P_1)Z.$$

If the first term were zero, then using the fact that L passes through P_1 , which by assumption may be written as $[t : 1 : 0]$, this implies that $F_Y(P_1) = 0$. Thus, either Z is tangent to C at P_1 or P_1 is not simple on C , which in both cases is a contradiction. Analogously, we obtain $F_Y \neq 0$.

Define the elements $x := X/Z$, $y := Y/Z$, $f_x := F_X(x, y, 1)$ and $f_y := F_Y(x, y, 1)$, these are all nonzero members of $K(C)$. Furthermore, we define $E_m := m \sum_{i=1}^n P_i - E$, where here, for every P_i , we identify it with the unique place on X corresponding to it.

The divisors, which can be represented as $\text{div}(G) - E$, where G ranges over all forms of degree $n - 3$, are all equivalent (since $\text{div}(G/G')$ is the difference of two such terms). Since if canonical divisors are, in fact, divisors (i.e. with finite support), they are all equivalent to one another, it suffices to show

$$\text{div}(dx) \stackrel{!}{=} \text{div}(Z^{n-3}) - E + \text{div}(f_y) = E_{n-3} + \text{div}(f_y) = E_{n-3} + \text{div}\left(\frac{F_Y(X, Y, Z)}{Z^{n-1}}\right).$$

In other words, it suffices to show

$$\text{div}(dx) - \text{div}(F_Y) = -2 \sum_{i=1}^n P_i - E. \quad (5)$$

Since $F(X, Y, Z)/Z^n = 0$ (in $K(C)$), we see that $f := F(x, y, 1) = 0$, so its differential is as well:

$$0 = df = f_x dx + f_y dy,$$

Therefore, since f_x is nonzero, we have the relation

$$dx = -\frac{f_y}{f_x} dy = -\frac{F_Y}{F_X} dy. \quad (6)$$

To show the equality in (5), we compare coefficients at every place $P \in X$. First, consider the coefficient at one of the P_i . Since $I(P_i; Z, F) = 1$ and $P_i \notin V(Y)$, it follows that the element $Z/Y = y^{-1}$ has order one in the local ring $\mathcal{O}_{P_i}(X)$, and hence is a uniformizing parameter. Furthermore, since

$$0 = d(1) = d\left(\frac{y}{y}\right) = yd\left(\frac{1}{y}\right) + \frac{1}{y}dy,$$

we see that $dy = -y^2 dy^{-1}$. Thus, from (6), we have

$$dx = \frac{F_Y}{F_X} y^2 dy^{-1}.$$

Accordingly, we may compute the coefficient of the left-hand side of (1) as:

$$\begin{aligned} \text{ord}_{P_i}(dx) - \text{ord}_{P_i}(F_Y) &= \text{ord}_{P_i}\left(\frac{F_Y}{F_X} y^2\right) - \text{ord}_{P_i}(F_Y) \\ &= \text{ord}_{P_i}(F_Y) + \text{ord}_{P_i}(y^2) - \text{ord}_{P_i}(F_X) - \text{ord}_{P_i}(F_Y) \\ &= -2 - \text{ord}_{P_i}(F_X) \\ &= -2. \end{aligned}$$

For the last equality, we used the fact that $F_X(P_i) \neq 0$: Indeed, using Euler's lemma, we write $nF = F_X \cdot X + F_Y \cdot Y + F_Z \cdot Z$. Therefore, if $F_X(P_i) = 0 = F(P_i)$, we must have that $F_Y(P_i) \cdot Y = 0$. Since $P_i \neq [1 : 0 : 0]$, this shows that $F_Y(P_i) = 0$ as well. This would imply that Z is tangent to C at P_i , which contradicts the assumptions about our coordinate system. Thus, the coefficients in (5) match at all points on $V(Z)$.

Now suppose P is some place on X centered at some $Q := [a : b : 1] \in C$. First, consider the case when $Y - Zb$ is tangent to C at Q . Then, by the choice of coordinates, Q is not a multiple point of C . Thus, $Y - Zb$ is the only tangent to C at Q ; in particular, we have that $X - Za$ intersects C transversely at Q . It follows that $(X - Za)/Z = x - a$ is a uniformizing parameter of the local ring $\mathcal{O}_P(X)$ and $F_Y(P) \neq 0$. Since $dx = d(x - a) = 1 \cdot d(x - a)$, we may compute the coefficient of the left-hand side of (5) at P as

$$\text{ord}_P(dx) - \text{ord}_P(F_Y) = 0.$$

This is also the coefficient of the right-hand side since the only affine points that have nonzero coefficients on the right-hand side are singular points.

If $Y - Zb$ is not tangent to C at Q , then the element $(Y - Zb)/Z = y - b$ is a uniformizing parameter of $\mathcal{O}_P(X)$. Since Y is not tangent, we can factor the lowest term $F_r(x, y, 1)$ of $F(x, y, 1)$ expanded at $[a : b : 1]$ as

$$F_r(x, y, 1) = \prod_{i=1}^r ((x - a) - \alpha_i(y - b))$$

(where we used the definition of a multiple point from Section 2). We find

$$\begin{aligned} f_x &= \frac{\partial}{\partial x} \prod_{i=1}^r ((x - a) - \alpha_i(y - b)) + \frac{\partial}{\partial x} F_{r+1}(x, y, 1) + \dots \\ &= \sum_{i=1}^r \prod_{j \neq i}^r ((x - a) - \alpha_j(y - b)) + (F_{r+1})_x + \dots, \end{aligned}$$

so the lowest term of f_x expanded about $[a : b : 1]$ has degree $r - 1$, thus f_x has order $r - 1$ there. Then, we can compute the coefficient of the left-hand side

once more as

$$\begin{aligned}
\text{ord}_P(dx) - \text{ord}_P(F_Y) &= \text{ord}_P\left(-\frac{f_y}{f_x}dy\right) - \text{ord}_P(F_Y) \\
&= \text{ord}_P\left(-\frac{f_y}{f_x}d(y-b)\right) - \text{ord}_P(F_Y) \\
&= \text{ord}_P(f_y) - \text{ord}_P(f_x) - \text{ord}_P(F_Y) \\
&= \text{ord}_P(f_y) - \text{ord}_P(f_x) - \text{ord}_P(f_y) \\
&= -(r-1),
\end{aligned}$$

which is the coefficient of the right-hand side. Since at any $P \in X$, the coefficients of the two divisors in 5 equate, they are the same divisor. $\square$

Corollary 7.2.6.1. *The degree of any canonical divisor is $2g-2$.*

Proof. Let C be any algebraic curve with nonsingular model X . We may choose a planar birational representative of X (hence also of C), which has only ordinary singularities and has degree $n \geq 3$, via quadratic transformations. Then we apply Theorem 7.2.6 to find that the canonical divisors have degree $n(n-3) - \deg(E)$ where E is the adjoint to C . By Lemma 5.1.9 this becomes

$$n(n-3) - \sum_{P \in C} m_P(m_P-1) = 2 \left(\frac{(n-1)(n-2)}{2} - \sum_{P \in C} \frac{m_P(m_P-1)}{2} \right) - 2.$$

By Theorem 6.2.1, the last term is $2g-2$. $\square$

Corollary 7.2.6.2. *Let W be a canonical divisor, then $l(W) \geq g$.*

Proof. Choose a plane curve of degree $n \geq 3$ birational to X , with only ordinary singularities. By Theorem 7.2.6 we know that $W := E_{n-3} := \text{div}(Z^{n-3}) - E$ is a canonical divisor. Denote by V_{n-3} the set of forms of degree $n-3$ which are adjoint to C . Using the proof of Theorem 6.2.1 we know that $l(W) = \dim_k(V_{n-3})$. By Lemma 3.3.1, we know this space to have dimension (not projective dimension) at least

$$\frac{(n-1)(n-2)}{2} - \sum_{P \in C} \frac{r_i(r_i-1)}{2} = g.$$

$\square$

8 Riemann-Roch Theorem

We have finally built the necessary machinery and are ready to prove the Riemann-Roch theorem. When D is a sufficiently large divisor, Corollary 6.1.2.1 already shows what the statement of the theorem should be. In its full generality, the theorem states:

Theorem 8.0.1 (Riemann-Roch Theorem). *Let X be a nonsingular algebraic curve of genus g , W be a canonical divisor on X , and D be a divisor on X . Then, the dimension of $L(D)$ may be calculated as*

$$l(D) = \deg(D) + 1 - g + l(W - D).$$

8.1 Proof of the Riemann-Roch Theorem

When D is a sufficiently large divisor, Theorem 5.2.2 guarantees that $l(W - D) = 0$, thus in this case, the theorem corresponds to Corollary 6.1.2.1. To prove the theorem, we only need to understand how $l(W - D)$ changes as we change D , specifically what happens as we add or subtract one point from D . Understanding how $l(D) - l(W - D)$ changes as we vary D will be the key point in the induction step in the proof of the Riemann-Roch theorem. This behavior is captured in the following theorem due to Noether.

Theorem 8.1.1 (Noether's Reduction Theorem). *Let X be a nonsingular projective curve and P be a point on X . Let D be a divisor on X with $l(D) > 0$. If W is a canonical divisor and $l(W - D - P) \neq l(W - D)$, then $l(D) = l(D + P)$.*

Proof. With desingularization, we may choose an algebraic curve $C \subseteq \mathbb{P}^2$ which has X as a nonsingular model and has only ordinary multiple points, on which P is a simple non-terrible point such that the defining (irreducible) form F has degree $n \geq 3$ (3.3.2). By Corollary 3.2.3.2, infinitely many lines intersect C transversely in $n := \deg(F)$ many points, such that none of these points are P . Hence, we may choose coordinates X, Y, Z of $\mathbb{P}^2$, such that $\text{div}(Z) = \sum_{i=1}^n P_i$, where the P_i are distinct and $P \neq P_i$ for all i (thus, these can be viewed as points on X).

Since each P_i corresponds to a unique point on X , define the divisor $E_m := m \sum_{i=1}^n P_i - E$ on X , where E is the adjoint divisor to C . Since E_{n-3} is a canonical divisor by Theorem 7.2.6 and the dimensions in the statement of our theorem depend only on linear equivalence, we may assume $W = E_{n-3}$ and $D \geq 0$ (replace D by $D + \text{div}(z)$, $z \in L(D) \setminus \{0\}$). Under these conditions we have $L(W - D) \subseteq L(E_{n-3})$. By the assumption of the theorem, we may choose $h \in L(W - D)$ and $h \notin L(W - D - P)$. Nevertheless, $h \in L(E_{n-3})$, thus we may write $h = \frac{G}{Z^{n-3}}$ where G is an adjoint of degree $n - 3$ due to Corollary 6.2.1.2. Then we have

$$\text{div}(h) = \text{div}(G) - \text{div}(Z^{n-3}) \geq D - W.$$

This shows $\text{div}(G) \geq E + D$; thus, we may write it as $\text{div}(G) = E + D + A$ for some effective divisor A . Since $h \notin L(W - D - P)$ we must also have $A \not\geq P$. Furthermore, since P is a simple, non-terrible point, we may choose a line L that intersects C in P transversely and in $n - 1$ distinct other simple points. Let B be the divisor consisting of these other $n - 1$ points (on X).

Suppose now we have some $f \in L(D + P)$, we will show that $f \in L(D)$ or equivalently, that $\text{div}(f) + D \geq 0$. Let $D' := \text{div}(f) + D$. By definition, D'

and D are equivalent divisors. Furthermore, since $f \in L(D + P)$, we know that $D' + P = \text{div}(f) + D + P \geq 0$, and that $D + P$ is an effective since D is. Since

$$\text{div}(LG) = (D + P) + E + (A + B),$$

we may apply the residue theorem (6.1.1) to find a form H of degree $n - 2$ with

$$\text{div}(H) = (D' + P) + E + (A + B).$$

The forms H and L have the $n - 1$ points of B in common, however since $\deg(H) = n - 2$, Bézout's theorem ensures that L is a component of H , so P lies on H as well. Therefore, D' must be effective and $f \in L(D)$ as desired. $\square$

Having proven Noether's reduction theorem, we can now prove the Riemann-Roch theorem. The idea of the proof is to use induction on $l(D)$ and to use Noether's reduction theorem as the bridge in our induction step.

Proof of the Riemann-Roch Theorem (8.0.1). First, we will prove the theorem for the case $l(W - D) = 0$. Then, we will pass to the more general case using Noether's reduction lemma.

Case 1: $l(W - D) = 0$. We proceed by induction over $l(D)$. If $l(D) = 0$, then Riemann's inequality ensures the two inequalities

$$\begin{aligned} 0 = l(D) &\geq \deg(D) + 1 - g, \\ 0 = l(W - D) &\geq \deg(W) - \deg(D) + 1 - g. \end{aligned}$$

Putting these two together yields

$$\begin{aligned} 0 = l(D) &\geq \deg(D) + 1 - g \\ &= \deg(D) + 1 - g + l(W - D) \\ &\geq \deg(D) + 1 - g + \deg(W) - \deg(D) + 1 - g \\ &= 2 - 2g + \deg(W) \\ &= 0. \end{aligned}$$

So $l(D) = \deg(D) + 1 - g + l(W - D)$ as desired.

If $l(D) = 1$, then D is equivalent to an effective divisor; so without loss of generality $D \geq 0$. From Corollary 7.2.6.2, we know $l(W) \geq g$, and from Theorem 5.2.2, we also have $l(W) \leq l(W - D) + \deg(D)$. Combining these two inequalities and using Riemann's inequality, we obtain:

$$\begin{aligned} l(D) &\geq \deg(D) + 1 - g \\ &= \deg(D) + 1 - g + l(W - D) \\ &\geq \deg(D) + 1 - g + l(W) - \deg(D) \\ &\geq 1 - g + g \\ &= l(D). \end{aligned}$$

So again, $l(D) = \deg(D) + 1 - g + l(W - D)$.

If $l(D) > 1$, we may choose some $P \in X$ such that $l(D - P) = l(D) - 1$, since we can simply choose some nonzero function $f \in l(D)$ and choose a point not in $(f)_0$, for then $f \notin L(D - P)$. Consequentially, from Noether's reduction lemma, it must hold true that $l(W - D - P) = l(W - D) = 0$. By induction and choice of P , the theorem holds for the divisor $D - P$, thus we obtain

$$\begin{aligned} l(D) &= l(D - P) + 1 \\ &= \deg(D - P) + 1 - g + l(W - D - P) + 1 \\ &= \deg(D) + 1 - g + l(W - D). \end{aligned}$$

Case 2: $l(W - D) > 0$. Since $l(W - D) > 0$, we have that this divisor must have a nonnegative degree, so we obtain $\deg(D) \leq 2g - 2$. Assume there exist divisors which make the theorem false. By Case 1, this can only occur when $l(W - D) > 0$; thus, all such divisors are bounded in degree by $2g - 2$. Hence, we may choose a maximal divisor D , making the theorem false. Additionally we choose some $P \in X$ such that $l(W - D - P) = l(W - D) - 1$. If $l(D) = 0$, the divisor $W - D$ satisfies the conditions of Case 1. Accordingly, we would then obtain

$$\begin{aligned} l(W - D) &= \deg(W - D) + 1 - g + l(W - W + D) \\ &= g - 1 - \deg(D) + l(D). \end{aligned}$$

However, this equation is equivalent to the theorem. If $l(D) > 0$, Noether's reduction lemma applies; therefore, $l(D + P) = l(D)$. By the maximality of D , the theorem holds true for the divisor $D + P$. consequentially we obtain

$$\begin{aligned} l(D) &= l(D + P) \\ &= \deg(D + P) + 1 - g + l(W - D - P) \\ &= \deg(D) + 1 - g + l(W - D), \end{aligned}$$

which contradicts the theorem not being true for D and thus finishes the proof.

□

9 Applications of the Riemann-Roch Theorem

In this section, we will delve into some of the implications of the Riemann-Roch theorem. Our first application of the theorem involves discussing linear series, a tool to map our nonsingular model X to different projective spaces. In Section 4, the nonsingular model X was seen to be projective by the Segre embedding.

However, the ambient space in which X resides in the construction in that section may be very large. Therefore, finding a systematic way to map the nonsingular model to different projective spaces might be convenient. Following this methodology, we will show that every genus one curve can be birationally embedded to a cubic in the projective plane, and every genus two curve can be viewed as a hyperelliptic curve.

As a second application, we will prove the Hurwitz formula. Suppose we are given a morphism $\Phi: X \rightarrow Y$ where Y is a variety we understand very well (for example $\mathbb{P}^1$). Hurwitz's formula provides a method of retrieving information about X , from Y and the morphism Φ . More explicitly, it will enable us to determine the genus of X using the genus of Y , the degree of Φ , and the ramified points of this covering (These notions will be defined later on).

9.1 Linear Series

We begin with the concept of a linear series.

Definition 9.1.1. Let D be a divisor and $V \subseteq L(D)$ be a linear subspace. The set of divisors obtained as

$$\{div(f) + D \mid f \in V\}$$

is called a linear series.

By the above definition, we see that the divisors appearing in the linear series are all effective and pairwise equivalent. We call a linear series complete if it contains all effective divisors equivalent to those in the linear series.

The main idea is that we fix a basis $f_0, \dots, f_r$ of V , after which we identify linear series with a projective space by the correspondence given by

$$div\left(\sum_{i=0}^r \lambda_i f_i\right) + D \leftrightarrow [\lambda_0 : \dots : \lambda_r] \in \mathbb{P}^r = \mathbb{P}(V).$$

More explicitly, any element of V is uniquely given by its corresponding coefficients λ_i , which in turn define a divisor in the linear series. Furthermore, two functions differ in a nonzero scalar multiple if, and only if, they define the same divisor; thus, this correspondence factors over forming the projective space $\mathbb{P}(V)$ over V .

Now consider the dual projective space $\mathbb{P}(V)^*$ to $\mathbb{P}(V)$ consisting of the hyperplanes of $\mathbb{P}(V)$. We will define a map from our curve X to $\mathbb{P}(V)^*$ as follows. First, assume that D is base-point-free, meaning there is no point P such that $V \subseteq L(D - P)$. Otherwise, we can replace D with the divisor $D - P$ and proceed until we eventually have this condition (we will assume this condition from now on). Under these assumptions, every P defines a unique hyperplane in V , given by

$$H_P := L(D - P) \cap V = \{f \in V \mid div(f) + D \geq P\}.$$

This is indeed a hyperplane, since $l(D - P) = l(D) - 1$ and $V \notin L(D - P)$. Projectivizing, we get a hyperplane in $\mathbb{P}(V)$, and thus a unique point in the dual space $\mathbb{P}(V)^*$. Given the above conditions, the following lemma shows that this map is a morphism of varieties.

Theorem 9.1.2. *The map Φ defined by*

$$\Phi: X \rightarrow \mathbb{P}(V)^*, P \mapsto H_P^*$$

is a morphism.

Proof. First, let $|D|$ denote the support of the divisor D . Then, $X \setminus |D|$ is an open subset of X . We claim that Φ is represented by a rational map defined on this open and is given by

$$\Phi': X \dashrightarrow \mathbb{P}^r, P \mapsto [f_0(P) : \dots : f_r(P)].$$

The map Φ' is a rational map, as it is given by global sections of $X \setminus |D|$, and it is well defined on this set since the functions f_i have no common zeros here and no poles. Since a rational map from a smooth curve to a projective curve is a morphism (Lemma 2.3.3), it remains to show that this map agrees with Φ . Suppose we write the hyperplane H_P as $H_P = V(a_i f_i)$, where the f_i are the chosen coordinates of $\mathbb{P}(V)$. Then Φ maps P to the point $[a_0 : \dots : a_r]$. The functions in H_P are precisely those functions that evaluate to zero at P (note that we need $P \notin |D|$ for this), so we may also calculate the above by:

$$\begin{aligned} 0 = f(P) &= \left(\sum_{i=0}^r \lambda_i f_i \right)(P) \\ &= \sum_{i=0}^r \lambda_i f_i(P). \end{aligned}$$

Since the λ_i are f' 's coordinates, we get that this hyperplane is defined by the $a_i = f_i(P)$. It remains to show that the extension of Φ' agrees with Φ on the support of D . Indeed, suppose Q is such a point occurring with multiplicity m in D and let $t \in \mathcal{O}_Q(X)$ be a uniformizing parameter. As before, the rational map defined on an open about Q as $P \mapsto [(t^m f_0)(P) : \dots : (t^m f_r)(P)]$ extends to a morphism which agrees with Φ' on a (dense) open, and so, everywhere. In particular since $[(t^m f_0)(Q) : \dots : (t^m f_r)(Q)] = H_Q^*$ we get the desired result. $\square$

We give the following lemma to understand the map Φ in more detail.

Lemma 9.1.3. *Let M be a hyperplane in $\mathbb{P}(V)^*$. The points of $\Phi^{-1}(M)$ correspond to the divisor $\text{div}(f) + D$ where f is the unique function in $\mathbb{P}(V)$ whose dual is M . Furthermore, the image $\Phi(X)$ is not contained in any hyperplane of $\mathbb{P}(V)^*$.*

Proof. By duality, we know that for a point P , we have $H_P^* \in M$ if and only if $M^* \in H_P$; since $M^* = f$ this is in turn equivalent to $\text{div}(f) + D \geq P$. This can only happen if P lies in the support of the divisor $\text{div}(f) + D$. It immediately follows that $\Phi(X)$ is not contained in any hyperplane since $\text{div}(f) + D$ has finite support for every f . $\square$

Remark. To make this more formal, we can associate to any hypersurface, an effective divisor in the same way as was done in Definition 5.1.

Corollary 9.1.3.1. *If $\mathbb{P}(V)$ is two-dimensional and Φ is injective, then the curve $C := \overline{\Phi(X)}$ is given by an irreducible form of degree $\deg(D)$.*

Proof. Let n be the degree of the defining form. By Corollary 3.2.3.2, we may find a line that intersects C transversally in n distinct points. By the previous lemma, these points correspond to the divisor $\text{div}(f) + D$, which is effective of degree $\deg(D)$. $\square$

The above construction gives a method to map a given algebraic curve to projective space. As the next lemma will show, any such map can be obtained in this way.

Lemma 9.1.4. *Let $\varphi: X \rightarrow \mathbb{P}^r$ be a morphism such that the image of X is not contained in any hyperplane of $\mathbb{P}^r$. Then the divisors corresponding to the sets $\varphi^{-1}(M)$ (counting multiplicities of intersections), where M ranges through all hyperplanes of $\mathbb{P}^r$, form a linear system.*

Proof. Let M_1 and M_2 be two such planes given by linear forms F_1, F_2 . Then $\varphi^{-1}(M_1)$ corresponds to $\text{div}(F_1) = \text{div}(F_2) + \text{div}(F_1/F_2)$. $\square$

Of particular interest is the case when the map Φ is an embedding. The Riemann-Roch theorem gives us a necessary criterion for when this is the case. Consider a complete linear series, i.e., $V = L(D)$. In this case, Φ becomes injective, provided that we choose D of a large enough degree.

Lemma 9.1.5. *If $\deg(D) \geq 2g + 1$, then Φ is injective, and in this case we call the space $L(D)$ very ample.*

Proof. suppose P, Q are two points on X with corresponding hyperplanes H_P, H_Q . Then $H_P = H_Q$ means $L(D - P) = L(D - Q)$. This in turn means that

$$L(D - P) = L(D - P - Q).$$

However, this is impossible since we may calculate dimensions using the Riemann-Roch theorem

$$\begin{aligned} l(D - P) &= \deg(D) - 1 + 1 - g + l(W - D + P) = \deg(D) - g, \\ l(D - P - Q) &= \deg(D) - 2 + 1 - g + l(W - D + P + Q) = \deg(D) - 1 - g. \end{aligned}$$

Here, we used Corollary 7.2.6.1 for the last step. $\square$

Using the theory of linear systems and the Riemann-Roch theorem, we get our first major application: The first step towards classifying genus one and genus two curves.

Theorem 9.1.6. *Every genus one curve is birationally equivalent to a smooth projective cubic plane curve (elliptic curve). Every genus two curve is birational to a hyper-elliptic curve. That is, a curve which admits a $2:1$ covering of $\mathbb{P}^1$.*

Proof. Suppose that C is a genus one curve and let X be its nonsingular model. Let P be any point on X and consider the divisor $D = 3P$. By the Riemann-Roch theorem, we may compute the dimension of $L(D)$ as:

$$l(D) = 3 + 1 - 1 + l(W - D) = 3.$$

Furthermore, D is base-point-free since an analogous calculation shows $l(3P - Q) = 2$ for any Q . In particular, by Theorem 9.1.2, we get a morphism $\Phi: X \rightarrow \mathbb{P}^2$, and, by Lemma 9.1.5 this morphism is injective and hence defines a birational equivalence. By Lemma 9.1.3.1, the closure $\overline{\Phi(X)}$ is given by an elliptic curve. The fact that this elliptic curve is smooth follows from the genus formula 6.2.1. Since Φ is a nonconstant morphism between projective curves, it must also be surjective, hence an isomorphism.

Suppose X has genus two, and suppose W is a canonical divisor on X . By Corollary 7.2.6.1 $\deg(W) = 2$. Hence, the Riemann-Roch theorem allows us to calculate the dimension of $L(W)$ as

$$l(W) = 2 + 1 - 2 + l(0) = 2.$$

Furthermore, $L(W)$ is base-point-free since by the Riemann-Roch theorem, we have $l(W - P) = l(P) = 1$ (here we use that $l(P) > 1$ would imply that X is rational). By Theorem 9.1.2, we get a nonconstant morphism from X to $\mathbb{P}^1$, which thus is surjective by Lemma 2.3.3. Finally, since the hyperplanes of $\mathbb{P}^1$ are simply points, Lemma 9.1 implies that this map is $2:1$ (counting multiplicities). $\square$

Remark. At the end of this section, we will clarify what we mean by “counting multiplicities” in our discussion about ramified points. In Section 10, we will demonstrate that the map for the genus two curve is also a topological ramified covering when $k = \mathbb{C}$.

Using a more subtle argument, we may improve the statement in Theorem 9.1.6 about genus one curves to the following theorem.

Theorem 9.1.7 (Weierstrass Formula). *Every smooth projective genus one curve is isomorphic to a smooth projective curve on which the canonical affine part is given by an equation of the form*

$$F = Y^2 + aXY + bY - X^3 + cX^2 + dX + e.$$

Proof. Let X be a smooth projective curve and W be a canonical divisor on X and $P \in X$. By Corollary 7.2.6 $\deg(W) = 0$, thus for all $n \geq 1$ the Riemann-Roch theorem states:

$$l(nP) = n + 1 - 1 + l(W - nP) = n.$$

This shows that $\{1\}$ is a basis for $L(P)$, and we may extend this to $\{1, x\}$ to obtain a basis of $L(2P)$. Since $x \notin L(P)$, x must have an order two pole at P . Once again, we can extend this basis to $\{1, x, y\}$ to obtain a basis of $L(3P)$. From this basis, we also see that $L(3P)$ is base-point-free, thus Theorem 9.1.2 ensures that we obtain a morphism $\Phi: X \rightarrow \mathbb{P}^2$ which is represented by the rational map $Q \mapsto [x(Q):y(Q):1]$. Furthermore, since $2g + 1 = 3$, Lemma 9.1.5 ensures that $L(D)$ is very ample. Since Φ is a nonconstant morphism between smooth projective curves, it is also surjective by Lemma 2.3.3 and hence an isomorphism.

Since $l(6P) = 6$ and the elements

$$\{1, x, x^2, x^3, y, y^2, xy\}$$

all lie in $L(6P)$, they must satisfy a nontrivial linear combination over k :

$$\lambda_1 + \lambda_2 x + \lambda_3 x^2 + \lambda_4 x^3 + \lambda_5 y + \lambda_6 y^2 + \lambda_7 xy = 0.$$

We observe that $\lambda_4 \neq 0$, since if $\lambda_4 = 0$, we would have

$$-\lambda_6 y^2 = \lambda_1 + \lambda_2 x + \lambda_3 x^2 + \lambda_5 y + \lambda_7 xy.$$

Comparing orders on the left and right at P , we see that $\lambda_6 = 0$ as well; however, we would then have the equation

$$-\lambda_7 xy = \lambda_1 + \lambda_2 x + \lambda_3 x^2 + \lambda_5 y.$$

This in turn implies that $\lambda_7 = 0$. Continuing this argument, we see that all coefficients vanish, which contradicts the choice of the λ_i . Similarly, we also find that $\lambda_6 \neq 0$. In particular, the points in the image of Φ on the affine $Z \neq 0$ all satisfy the equation:

$$F = \lambda_6 Y^2 + a' XY + b' Y - \lambda_4 X^3 + c' X^2 + d' X + e',$$

for nonzero λ_1, λ_2 . Note that the form defining the curve on all of $\mathbb{P}^2$ is given by homogenizing F . The polynomial F must be irreducible and define a smooth curve by the genus formula in 6.2.1, since any other case would not allow X to have genus one. Finally, the projective change of coordinates given by

$$T: \mathbb{P}^2 \rightarrow \mathbb{P}^2, [X:Y:Z] \mapsto \left[\frac{X}{\lambda_4^{1/3}} : \frac{Y}{\lambda_6^{1/3}} : Z \right],$$

transforms F to

$$F^T = F \circ T = Y^2 + aXY + bY - X^3 + cX^2 + dX + e$$

for suitable coefficients $\{a, b, c, d, e\}$. □

9.2 Hurwitz Formula

As a final application of the Riemann-Roch theorem, we prove the Hurwitz formula, which will be the key theorem we need in Section 10 to prove the equivalence of the algebraic genus defined here and the usual topological genus of a closed surface. We are faced with the following problem: Given a (surjective) morphism $f: X \rightarrow Y$ of smooth projective curves, we would like a method for determining the genus of X from that of Y and properties of f . One way to solve this problem is to observe how differentials on Y relate to differentials on X and use the fact that the degree of a differential can be expressed in terms of the genus by Corollary 7.2.6. This heuristic is made precise in our proof of the Hurwitz formula. Before we do this, we first need to prove two preliminary results. Note first that if $f: X \rightarrow Y$ is a surjective morphism, the induced field homomorphism $\tilde{f}: K(Y) \rightarrow K(X)$ of function fields is injective (Lemma 2.3.3). Therefore, we can view $K(Y)$ as a subfield of $K(X)$. Under these conditions, we have made the following definition adapted from [9].

Definition 9.2.1. Suppose $f: X \rightarrow Y$ is a nonconstant (hence surjective) morphism of smooth projective curves. The degree of f is defined to be the degree of the field extension $[K(X):K(Y)]$. If P is a point and X and $f(P) = Q \in Y$, let $t \in \mathcal{O}_Q(Y)$ be a uniformizing parameter. The ramification index of f at P is defined as $e_f(P) := \text{ord}_P(t)$. We say that the ramification (index) is tame if $\text{char}(k) = 0$ or $\text{char}(k) = p$ and p does not divide $e_f(P)$; otherwise, the ramification is called wild.

Remark. The ramification index is independent of the choice of uniformizing parameter. Indeed, since f is a morphism, the local ring $\mathcal{O}_Q(Y)$ is contained in $\mathcal{O}_P(X)$. Therefore, units in $\mathcal{O}_Q(Y)$ are also units in $\mathcal{O}_P(X)$. Since any two uniformizing parameters differ in a unit in $\mathcal{O}_Q(Y)$, the result follows.

Theorem 9.2.2. Suppose $f: X \rightarrow Y$ is a morphism of smooth projective curves of degree n . Let Q be a point on Y . Then, the divisor $D := \sum_{P \mapsto Q} e_f(P)P$ is an effective divisor of degree n .

Remark. The idea of the following proof has been adapted from Andrew V. Sutherland's lecture notes to the course Arithmetic Geometry, which can readily be found at [8].

Proof. Let $t \in \mathcal{O}_Q(Y)$ be a uniformizing parameter; $P_1, \dots, P_m$, be the points in X mapping to Q with respective local rings $\mathcal{O}_{P_i}$; and let $\mathcal{O} := \bigcap_{i=1}^m \mathcal{O}_{P_i}$. Passing to an affine curve where the P_i are smooth points, we find that we can choose $t_1, \dots, t_m \in \mathcal{O}$ such that $\text{ord}_{P_i}(t_j) = \delta_{ij}$ (in the affine case we may simply take lines L_i which intersect the curve transversally in P_i and pass through none of the other P_j and then divide each of these by a line which passes through none of the points). In particular, all t_i are units in $\mathcal{O}$. Using this notation, we may write $t = ut_1^{e(P_1)} \dots t_m^{e(P_m)}$, with u being a unit in $\mathcal{O}$. Since the ideals (t_i) are

pairwise coprime, the Chinese remainder theorem ensures that

$$\mathcal{O}/(t) \simeq \prod_{i=1}^m \mathcal{O}/(t_i^{n_i}), \text{ (where } n_i = e(P_i)\text{)}.$$

We claim that the dimension of $\mathcal{O}/(t)$ as a k -vector space is both equal to $\deg(D)$ and to $\deg(f)$, which would prove the theorem. By power series expansion, any element $x \in \mathcal{O}_{P_i}$ may be written as a unique sum

$$x = \lambda_0 + t_i \lambda_1 + \cdots + \lambda_{e(P_i)-1} t_i^{e(P_i)-1} + g t^{e(P_i)},$$

where $\lambda_i \in k$ and $g \in \mathcal{O}_{P_i}$. In particular, those elements of $\mathcal{O}$ can be uniquely represented in the above form. From this expansion, we also see that for $x \in \mathcal{O}_{P_i}$, we have $x \in \mathcal{O}$ if and only if $g \in \mathcal{O}$, since the first terms in the above sum all lie in $\mathcal{O}$ (using $t_i \in \mathcal{O}$). Taking the residues of $1, t_i, \dots, t_i^{e(P_i)-1}$ in $\mathcal{O}/(t_i^n)$, it follows that $\dim_k(\mathcal{O}/(t_i^n)) = e(P_i)$, which in turn implies $\dim_k(\prod_{i=1}^m \mathcal{O}/(t_i^{n_i})) = \deg(D)$.

The ring $\mathcal{O}$ is also a finitely generated module over $\mathcal{O}_Q(Y)$ via the homomorphism $\tilde{f}$, and we claim that the rank of this module is equal to n . Indeed, $K(Y)$ is the field of fractions of $\mathcal{O}$ and $K(X)$ is the field of fractions of $\mathcal{O}_Q(Y)$. Thus, given a basis $x_1, \dots, x_n$ of $K(X)$ over $K(Y)$, we may assume these to lie in $\mathcal{O}$ by clearing out the denominators. These elements must then be $\mathcal{O}_Q(Y)$ -linearly independent as well. Conversely, any set of elements in $\mathcal{O}$ which are linearly independent over $\mathcal{O}_Q(Y)$ must also be independent over $K(Y)$, since if we had a nontrivial linear relation, this would induce a nontrivial linear combination over $\mathcal{O}_Q(Y)$ by clearing out the denominators of the appearing coefficients. Finally, to complete the proof, we have that (t) is the unique maximal ideal of $\mathcal{O}_Q(Y)$. Therefore, by Nakayama's lemma $\mathcal{O}/(t)$ is a $\mathcal{O}_Q(Y)/(t) \simeq k$ vector space of dimension n . $\square$

To finish our preparations for the Hurwitz formula, we need to understand how differentials on Y are related to differentials on X . If

$$\Phi: K(X) \mapsto \Omega_k(K(X))$$

is the differential map, then we get a derivation on $K(Y)$ via the map $\Phi \circ \tilde{f}$. Thus, from the universal property in Lemma 7.1.3 we get a natural morphism $\tilde{\Phi}: \Omega_k(K(Y)) \rightarrow \Omega_k(K(X))$ such that the following diagram commutes:

$$\begin{array}{ccc} K(Y) & \xrightarrow{\tilde{f}} & k(X) \xrightarrow{\Phi} \Omega_k(K(X)) \\ \downarrow d & & \nearrow \tilde{\Phi} \\ \Omega_k(K(Y)) & & \end{array}$$

So, we may view the differentials on Y as differentials on X via the above diagram. Going a step further, for a fixed differential $\omega \in \Omega_k(K(Y))$, we can ask how its divisor on Y compares to that of its divisor on X . The next lemma answers this question for differential forms of the form dt , where t is a uniformizing parameter at a point on Y . Using the above notation, we get the following lemma.

Lemma 9.2.3. *Suppose $Q \in Y$, and let t be a uniformizing parameter of Y at Q . Furthermore, assume that f only has tame ramifications. Then for each point $P \in X$, mapping to Q , we have $\text{ord}_P(dt) = e(P) - 1$.*

Proof. Let s be a uniformizing parameter of X at P . Then by definition we have $t = v^{e(P)} \cdot u$ where $\text{ord}_P(u) = 0$. By Computing differentials, we find

$$\begin{aligned} dt &= e(P)v^{e(P)-1}udv + v^{e(P)}du \\ &= e(P)v^{e(P)-1}udv + v^{e(P)}\frac{du}{dv}dv \\ &= \left(e(P)u + v\frac{du}{dv}\right)v^{e(P)-1}dv \end{aligned}$$

which shows that $\frac{dt}{dv} = \left(e(P)u + v\frac{du}{dv}\right)v^{e(P)-1}$. By Theorem 7.2.2 $\frac{du}{dv} \in \mathcal{O}_P$ and since $e(P)$ is a tame ramification, we see that $\text{ord}_P\left(e(P)u + v\frac{du}{dv}\right) = 0$, which implies the result. $\square$

With these preparations, we can prove the main result needed for Section 10.

Theorem 9.2.4 (Hurwitz Formula). *Suppose $f: X \rightarrow Y$ is a nonconstant morphism of smooth projective curves such that f has only tame ramification indices. Let g_X, g_Y be the genres of X and Y , respectively. Then the following relation holds:*

$$2g_X - 2 = (2g_Y - 2)\deg(f) + \sum_{P \in X} (e(P) - 1).$$

Proof. Let $W = \text{div}_Y(\omega)$ be a canonical divisor on Y . By Corollary 7.2.6.1 we know that $\deg(W) = 2g_Y - 2$. Suppose Q is any point on Y and let n_Q be the coefficient with which it appears in W . Then we may choose a uniformizing parameter t at Q such that $\omega = g \cdot dt$ for some $g \in k(Y)$ with $\text{ord}_Q(g) = n_Q$. In particular, if we view this in $\Omega_k(K(X))$ we obtain

$$\omega = \left(g \cdot \frac{dt}{dv}\right)dv,$$

where dv is a uniformizing parameter at one of the points P mapping to Q . Since g has order n_Q at Q , g must have order $n_Q \cdot e(P)$ at P . Thus, by the previous lemma, ω must have order $n_Q \cdot e(P) + e(P) - 1$ at any such P . Using this information we calculate the degree of the divisor $\text{div}_X(\omega)$ on X as

$$\begin{aligned} \deg(\text{div}_X(\omega)) &= \sum_{Q \in Y} \sum_{P \mapsto Q} (n_Q \cdot e(P) + e(P) - 1) \\ &= \sum_{Q \in Y} \sum_{P \mapsto Q} (n_Q \cdot e(P)) + \sum_{P \in X} (e(P) - 1) \\ &= \sum_{Q \in Y} n_Q \left(\sum_{P \mapsto Q} e(P) \right) + \sum_{P \in X} (e(P) - 1). \end{aligned}$$

Using Theorem 9.2.2, we may further simplify this to

$$\begin{aligned}
\sum_{Q \in Y} n_Q \left(\sum_{P \mapsto Q} e(P) \right) + \sum_{P \in X} (e(P) - 1) &= \sum_{Q \in Y} n_Q \deg(f) + \sum_{P \in X} (e(P) - 1) \\
&= \deg(f) \sum_{Q \in Y} n_Q + \sum_{P \in X} (e(P) - 1) \\
&= \deg(f)(2g_Y - 2) + \sum_{P \in X} (e(P) - 1).
\end{aligned}$$

Now $\text{div}_X(\omega)$ is a differential on X , thus by Corollary 7.2.6.1, it has degree $2g_X - 2$. By equating these two, the result follows. $\square$

Remark. Since canonical divisors have finite support, as proven in Theorem 7.2.6, we consequentially obtain that $e(P) > 1$ happens at only finitely many points on X . These points are called ramification points.

10 The Topological Genus

As previously announced, this chapter aims to connect the algebraic notions used in this thesis with their analytic and topological counterparts. We will consider an algebraic curve X , which lies in $\mathbb{P}\mathbb{C}^n$. Furthermore, we will only view smooth closed curves since the usual genus of a Riemann surface is defined only for such curves. We will use some standard results from complex analysis and topology without providing their proofs.

10.1 Analytic Structure

We first describe how to give our algebraic curve the structure of a Riemann surface by describing a topology and an atlas. For a topology, take the induced subspace topology of $\mathbb{P}\mathbb{C}^n$ (here we mean the topology that this space carries as an identification space of $\mathbb{C}^{n+1} \setminus \{0\}$). First note that for any form F the, hypersurface $V(F) = F^{-1}(\{0\})$, is closed in $\mathbb{P}\mathbb{C}^n$ as the preimage of a closed set. Since X is given by an intersection of such hypersurfaces, it is a closed subset of a compact space and hence compact as well. Now let $P \in X$ be any point. We will describe a chart around P as follows. Choose some uniformizing parameter $t \in \mathcal{O}_P$. Since t does not have a pole at P and t has only finitely many poles and zeros on X , there exists an open neighborhood U_P of P on X such that the map $Q \mapsto t(Q)$ is well defined to $\mathbb{C}$ and has only one zero at P . These maps can be viewed as charts as seen in the following theorem, the proof of which may be found in chapter III of [7].

Theorem 10.1.1. *The maps $\varphi_P: D_P \rightarrow \mathbb{C}, Q \mapsto t(Q)$ give homeomorphisms from sufficiently small open disks D_P around P to open disks of $0 \in \mathbb{C}$. Furthermore for two points P, Q the transition maps $\varphi_Q \circ \varphi_P^{-1}$ are analytic.*

In particular, Theorem 10.1.1 ensures that X is a Riemann surface in a natural way.

The functions in $K(X)$ are given by quotients of homogeneous functions (since we assumed X was embedded in a projective space). They are all meromorphic; that is, for $f \in K(X)$ the maps $f \circ \varphi_P^{-1}$ are meromorphic on $\mathbb{C}$. From the representation of f as a rational function, we also see that the analytic notions of order and poles coincide with the algebraic ones.

We will now show that the Riemann surface is a closed surface. We already know it is compact, so we are left to show that X is connected.

Lemma 10.1.2. *The algebraic curve X is a connected Riemann surface.*

Proof. Suppose C is any connected component of X and let $P \in C$. Suppose C is not the entire curve. By the Riemann-Roch theorem, for sufficiently large n , there exists an $f \in L(nP)$ with a pole at P , which is holomorphic everywhere else. In particular, f is holomorphic on any component other than C . Any such component is again a compact Riemann surface, thus f exhibits a maximum there; by the maximum modulus principle of complex analysis, we have that $f = \lambda$ is constant on any such component. Then, the zero set of $f - \lambda$ is an algebraic subset of X , which contains infinitely many points. Since X is irreducible and of dimension one, this set is all of X , which contradicts f having a pole at P . $\square$

The classification theorem of closed surfaces (compact and without boundary) in topology ensures that every closed surface is topologically given by a sphere with either a finite number of handles attached or a sphere with a finite number of disks removed and Möbius bands glued in, with the latter case only occurring for nonorientable surfaces. By Lemma 10.1.2, X is a compact Riemann surface, in particular, a closed surface. Hence, up to homeomorphism, X is given by the number of handles attached since analytic $\mathbb{C}$ manifolds are always orientable.

We will now compute the topological genus of our curve by giving a suitable triangulation. Let K be the function field of X , and suppose z is a nonconstant function in K . By Theorem 6.1.2, we know that the divisor of zeros $(z)_0$ is of degree $n = [K:\mathbb{C}(z)]$. Therefore, the divisor of poles $D := (z)_\infty$ must be as well. Furthermore, the set $\{1, z\}$ spans a two-dimensional base-point-free sub-vector space V of $L(D)$. So by Theorem 9.1.2, we get a nonconstant morphism $f_z: X \rightarrow \mathbb{P}^1$, which is represented at all but finitely many points as $P \mapsto [1:z(P)]$. Since z is a rational function, this map is a continuous map to the Riemann sphere. By Theorem 9.2.2, and the fact that any morphism between smooth curves has only finitely many ramification points as shown in the proof of the Hurwitz formula 9.2.4, we see that the preimage of all points except those corresponding to ramification points consists of exactly n distinct points. We need the following lemma to conclude that our map f_z is a ramified n -sheeted covering in the sense of complex analysis.

Lemma 10.1.3. *Suppose $f_z: X \rightarrow \mathbb{P}^1$ is as above. Then f_z is a ramified (analytic) covering, and a point $P \in X$ is ramified if and only if it is a branch point of this cover. Furthermore, the order of the branching at P corresponds to the algebraic ramification index.*

Proof. Let Q be a point in $\mathbb{P}\mathbb{C}^1$ and P be a point in X mapping to P . Let X and Y be homogeneous coordinates for $\mathbb{P}\mathbb{C}^1$. Without loss of generality, we may assume that Q is not on $V(X)$. Identifying the affine $X \neq 0$ with $\mathbb{C}$, we find a uniformizing parameter of $\mathcal{O}_Q(\mathbb{P}\mathbb{C}^1)$ given by $z - a$ where $Q = [1 : a]$ and $z = Y/X$. Let t be a uniformizing parameter of $\mathcal{O}_P(X)$. Then, by the definition of the algebraic ramification number, we get that $z - a = t^{e(P)} \cdot u$, where u is some unit in $\mathcal{O}_P(X)$. Since $t^{e(P)} \cdot u$ is holomorphic and has zero of order $e(P)$ at P . From complex analysis, it is known that we may then represent it as $h^{e(P)}$ where h is a holomorphic function that has a simple zero at P . Since h has a simple zero at P , it gives a biholomorphism from a sufficiently small disk around P to one around 0 (inverse mapping theorem). Therefore the map $h^{e(P)} = (\cdot)^{e(P)} \circ h$ must give an $e(P) : 1$ (analytic) ramified covering with branch point P of a disk around zero. Surely, translating this map: $h^{e(P)} + a$ then gives a ramified covering of a disk around Q . Since $h^{e(P)} + a = t^{e(P)} \cdot u + a = z$ the result follows for z . $\square$

10.2 Euler Characteristic and Genus

Instead of calculating the topological genus of X , we first calculate its Euler characteristic. The main idea in doing this is to triangulate $\mathbb{P}\mathbb{C}^1$ in such a way that the map f_z induces a triangulation on X , which is easy to work with, where afterward we calculate $\chi = V - E + F$.

Theorem 10.2.1. *Let X be a smooth projective curve over $\mathbb{C}$ and z be a non-constant element of X 's function field. Furthermore, let f_z be the morphism that z induces onto $\mathbb{P}\mathbb{C}^1$ and n the degree of the covering. Then the Euler characteristic χ of X is given by:*

$$\chi = 2n - \sum_{P \in X} (e(P) - 1).$$

Proof. We start with an arbitrary triangulation of $\mathbb{P}\mathbb{C}^1$. By the remark at the end of Section 9, there are only finitely many points in $\mathbb{P}\mathbb{C}^1$ which lie under ramified points of X . Thus, if needed, we add extra vertices, edges, and faces to this triangulation so that every point in $\mathbb{P}\mathbb{C}^1$, which lies under a ramified point, becomes a vertex of this triangulation. Using barycentric subdivision, we may assume that a triangle contains at most one such vertex. By the above theorem, for every $P \in X$, there exists an open disk D_P on X such that f_z gives an $e(P) : 1$ map from an open disk around P to an open disk D_Q with $Q := f_z(P)$. Taking for each $Q \in \mathbb{P}\mathbb{C}^1$ the smallest disk induced from the points in X , which lay above it, we get an open cover of $\mathbb{P}\mathbb{C}^1$, which admits a finite subcover. Using barycentric subdivision, we may thus assume that every triangle in our triangulation, which does not contain a ramified point as a vertex, is completely contained in such a disk D_Q , where Q is a regular point. Furthermore, since there are only finitely many points lying under ramified points, we may subdivide the triangles with vertices at ramified points such that they are completely contained in the smallest of the disks induced by points above the ramified point.

Each triangle with no ramified vertices then lifts uniquely to n triangles in X since they are completely contained in the disk around some point that does not correspond to a ramified point. On the other hand, consider a triangle with vertex Q that lies under ramified point $P \in X$. By the choice of triangulation, there is a disk D_P about P and a disk D_Q about Q containing the triangle such that f_z restricted to D_P is a $e(P):1$ map with unique branchpoint P . By the proof of the previous theorem, we also know that f looks like $t^{e(P)} + a$ where t is a biholomorphism to a disk around zero, and adding a translates this to a disk around Q . The map $z \mapsto z^{e(P)} + a$ collapses $e(P)$ different triangles, differing by roots of unity, into one triangle (If needed, subdivide the triangle sufficiently small). This is illustrated in the diagram below.

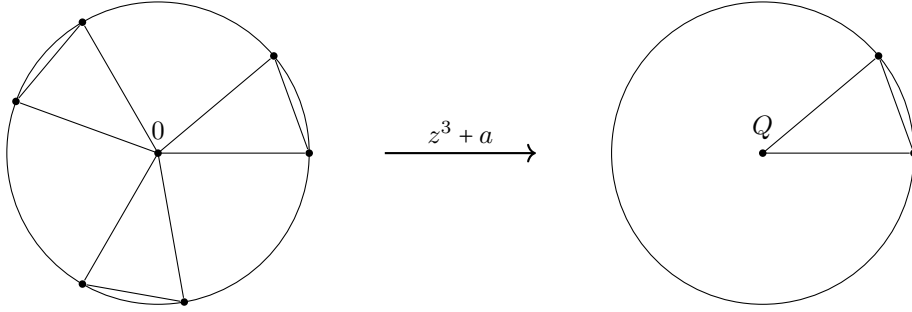


Figure 5: Lifting a triangulation to a ramified covering

Since t is a biholomorphism, it follows that the triangle lifts to $e(P)$ different triangles in D_P alone. In particular, in total, the triangle lifts to

$$\sum_{P_i \mapsto Q} e(P_i) = n$$

different triangles in X , where we used Lemma 9.2.2 to obtain that the sum evaluates to n .

Let V, E , and F denote the number of vertices, triangles, and faces used in our triangulation of $\mathbb{P}\mathbb{C}^1 \simeq \mathbb{S}^2$. From topology, we know that $\mathbb{S}^2$ has Euler characteristic two since it is homeomorphic to a tetrahedron. Thus we have

$$2 = V - E + F.$$

Suppose V', E' , and F' are the respective numbers of vertices, edges, and faces of the lifted triangulation to X . Owing to the previous discussion, we have

$$E' = nE$$

$$F' = nF.$$

Write $V = V_1 \cup V_2$ where V_1 is the set of vertices not lying under ramified points and V_2 is the remaining set of points. For each point, $Q \in V_2$, denote by n_Q the

number of points in X that map to Q . We may then deduce

$$\begin{aligned}
V' &= n |V_1| + \sum_{Q \in V_2} n_Q \\
&= nV - n |V_2| + \sum_{Q \in V_2} n_Q \\
&= nV + \sum_{Q \in V_2} (n_Q - n) \\
&= nV + \sum_{Q \in V_2} \sum_{P \mapsto Q} (1 - e(P)), & (\text{Lemma 9.2.2}) \\
&= nV + \sum_{Q \in V} \sum_{P \mapsto Q} (1 - e(P)), \\
&= nV - \sum_{P \in X} (e(P) - 1).
\end{aligned}$$

Finally, we find the Euler characteristic χ_X of X as

$$\begin{aligned}
\chi_X &= V' - E' + F' \\
&= nV - nE + nF - \sum_{P \in X} (e(P) - 1) \\
&= 2n - \sum_{P \in X} (e(P) - 1).
\end{aligned}$$

□

We can now provide a short proof of the main result of this section.

Theorem 10.2.2. *Let X be a smooth projective algebraic curve over $\mathbb{C}$. Then, the algebraic genus is equal to the topological genus.*

Proof. Let z be a nonconstant function in the function field of x and let n be the degree of the branched cover of $\mathbb{P}^1$ that f_z defines. Since we know that $\mathbb{P}^1$ has algebraic genus zero, the Hurwitz formula in Theorem 9.2.4 ensures that the algebraic genus g^* of X satisfies

$$2g^* - 2 = -2n + \sum_{P \in X} (e(P) - 1).$$

On the other hand, if χ is the Euler characteristic of X , we know from topology that the topological genus g of X satisfies $2 - 2g = \chi$. Theorem 10.2.1 then implies

$$2g - 2 = -\chi = -2n + \sum_{P \in X} (e(P) - 1).$$

Thus, we find $2g^* - 2 = 2g - 2$.

□

References

- [1] Riemann, Bernhard (1857) *Theorie der Abel'schen Funktionen* [Journal für die reine und angewandte Mathematik, Bd. 54. S. 101–155. 1857.]

- [2] Roch, Gustav (1865) *Ueber die Anzahl der willkürlichen Constanten in algebraischen Functionen*. [Journal für die reine und angewandte Mathematik 64: 372–376]
- [3] Dedekind, Richard and Weber, Heinrich (1882) *Theorie der algebraischen Functionen einer Veränderlichen* [J. für Math., 92, 181-290]
- [4] Brill, Alexander and Noether, Max. (1874) *Über die algebraischen Functionen und ihre Anwendung in der Geometrie* [Math. Ann., 7, 269-310]
- [5] Fulton, William (1989) *Algebraic Curves: An Introduction to Algebraic Geometry*
- [6] Kendig, Keith (1977) *Elementary Algebraic Geometry*
- [7] Lang, Serge (1982) *Intorduction to abelian functions*
- [8] Sutherland, Andrew (2013) *Introduction to Arithmetic Geometry MIT OpenCourseWare*
- [9] Hartshorne, Robin (1977) *Algebraic Geometry*
- [10] Blowup of a Curve with Ordinary Singularities (accessed on 11.8.2024) <https://www.math.purdue.edu/~arapura/graph/nodal.html>
- [11] Grammarly (accessed 2024) [Used for grammar corrections.] <https://www.grammarly.com/grammar>